

A true vertex model for epithelial mechanics in plexus2: force balance, explicit T1, and what the Voronoi route could not do

Notes from a plexus2 prototype of tyssue (DamCB)

July 2026 — *living document, updated as the prototype progresses*

Abstract

This is the companion to the Turing-vertex prototype. There we modelled epithelial mechanics as a *Self-Propelled Voronoi* (SPV) tissue — each cell a point, the polygon its Voronoi cell, the mesh re-tessellated every tick so neighbour exchanges (T1) were implicit. That report closed on a clear diagnosis: clean *tubulation* was gated not on chemistry but on two mechanics ingredients the Voronoi route structurally lacks — (i) *force-balance iteration* and (ii) an *explicit reconnection* (T1) operator. Here we build the sibling implementation: a *true vertex model* (AVM) after **tyssue**, whose degrees of freedom are the *vertices* of a half-edge mesh, in which those two ingredients are native. The same epithelial shape-energy contract, a genuinely different numerical realisation — the “one contract, many implementations” thesis of **plexus2** §5, made concrete. This document records the model-deciphering as it happens; findings are added as each stage lands. **[Results sections are filled in incrementally — see the dated field notes.]**

1 Two implementations of one contract

Both prototypes realise the same biological operator — an epithelial *mechanical interaction* whose energy is a per-cell area/perimeter shape energy

$$E = \sum_f \left[K_A (A_f - A_f^0)^2 + K_P (P_f - P_f^0)^2 \right] + \frac{1}{2} \Gamma \sum_f P_f^2 + \Lambda \sum_e \ell_e / 2,$$

with a target shape index $p_0 = P^0 / \sqrt{A^0}$ tuning a rigidity transition at $p_0^* \approx 3.81$. They differ only in *how* the polygons and their motion are represented:

	SPV (Turing_vertex)	AVM (this prototype, tyssue)
degrees of freedom	cell <i>centres</i> (points)	mesh <i>vertices</i> (half-edge mesh)
polygon	Voronoi cell of the centres	explicit face = ring of shared vertices
adjacency / T1	re-tessellate each tick $\Rightarrow$ <i>implicit</i>	<i>explicit</i> local edge collapse + split
force balance	overdamped Euler steps	gradient descent to residual (quasistatic)
cost	Delaunay per tick (scipy)	one scatter-add + one autograd pass

The AVM is the missing sibling: it supplies force balance and explicit T1, which is exactly what the SPV report identified as the tubulation bottleneck. Concretely, both are registered `plexus2` operators acting on a `set`; the SPV acts on a `cell` set, the AVM on a `vertex` set with the faces carried as a half-edge table.

The original (Okuda) is a 3D vertex model — this is the 2D rung of that ladder. Okuda *et al.* is a *true 3D* vertex model: cells are polyhedra sharing polygonal faces, the DOF are the junction vertices $\mathbf{r}_i$ (overdamped $\eta_i(\dot{\mathbf{r}}_i - \mathbf{V}_{fi}) = -\nabla_i U$), the energy $U = \sum_j \frac{1}{2} k_v (v_j - v_j^{\text{eq}})^2 + \kappa_s s_j$ is over each cell’s polyhedral *volume* and *surface*, the balance is iterated to residual $< 10^{-5}$ every Δt_v , neighbour changes are explicit *reversible network reconnection* (RNR / T1) every Δt_r , and division inserts a new face. The prototype here is the *2D* rung of that same ladder — vertices, force balance, explicit T1 — with an area/perimeter energy standing in for volume/surface. That is deliberate: 2D is where T1 is a clean edge flip, so we validate force balance and reconnection there first. `tyssue` itself carries the 3D rungs (`Monolayer/Bulk`: `cell.volume` elasticity, the IH/HI 3D reconnections, `extrude` apical $\rightarrow$ 3D), all catalogued in the extraction (§4) — so the path to Okuda’s elongated *tubes* (rather than the SPV prototype’s differential-growth *lobes*) is to lift this 2D core to a monolayer once the 2D topology operators are validated.

2 The model in one paragraph

The mechanical set is the `vertex` (state `pos`); the tissue topology is a *half-edge table* (`srce`, `trgt`, `face`) ordered counter-clockwise around each face, stashed on the vertex `Level` exactly as the SPV prototype stashed its Voronoi rings. Per-face targets A^0, P^0 and liveness live alongside it. `mesh_seed` bootstraps a clean honeycomb by taking the Voronoi tessellation of a triangular lattice *once* (outer rings dropped, their shared vertices pinned $\rightarrow$ a fixed-border patch), then the AVM takes over. `shape_energy` computes every face’s area and perimeter with a single `index_add` over the half-edge table, forms the shape energy, and takes the force $-\nabla E$ on the vertices by one autograd pass; an inner loop of `relax_iters` gradient steps per tick returns the net displacement as a velocity, so each rendered frame is (near-)force-balanced — the quasistatic force balance the original does with `scipy` L-BFGS, here vectorised and differentiable. Boundary (pinned) vertices are held fixed.

A half-edge primer. A *half-edge* is a *directed* edge $A \rightarrow B$ that belongs to exactly one face; the real edge shared by cells L and R is stored as two opposite half-edges. Each is three integers (`srce`, `trgt`, `face`):

```

    cell L    (interior on the left)
... -> A -----> B -> ...    half-edge  A->B ,   face = L
... <- A <----- B <- ...    half-edge  B->A ,   face = R
    cell R
E_srce[k] = A      E_trgt[k] = B      E_face[k] = L    (one row per half-edge)

```

A face is the ring of its half-edges in CCW order, so orientation — and a positive shoelace area — is automatic; the two half-edges of a junction name its two neighbour cells, which is exactly what a T1 needs (grab $A \rightarrow B$ in L and $B \rightarrow A$ in R , collapse, re-split); and per-face area/perimeter are a single `index.add` keyed by `E_face` (no per-cell loop, so the mechanics vectorise). In `plexus2` terms `srce/trgt/face` are three index-buffer *maps* on the set: `srce/trgt` are edge→vertex incidence, `face` is edge→cell containment; gathering a cell value onto its half-edges is a *broadcast*, reducing half-edges to a cell is an *aggregate*. Line tension is $\Lambda\ell/2$ *per half-edge*, so the two halves sum to $\Lambda\ell$ per real edge — tyssue’s half-edge convention.

3 Design decisions deciphered so far

(These are the modelling choices that were not obvious from the paper and cost thought.)

Vertices, not cells, are the set. The natural `plexus2` temptation is to keep the `cell` set of the SPV prototype and bolt a mesh on. That is wrong: the AVM *integrates the vertices*. Cells share vertices, so a cell is not an independent particle — it is a face, a derived readout over a ring of the vertex set. The clean mapping is therefore a **vertex** set (the DOF) plus the half-edge table as its topology map; faces never get their own integrated `Level`. Area/perimeter are *aggregate* readouts (vertices→face), the force is their gradient back on the vertices — a genuine cross-scale (aggregate/broadcast) pair, richer than the single-set SPV.

Bootstrap the mesh from a Voronoi, then leave Voronoi behind. Hand-building a periodic honeycomb half-edge mesh with correct orientation is fiddly and error-prone. Taking the Voronoi tessellation of a jittered triangular lattice *once* gives a valid honeycomb for free (`scipy` already de-duplicates the Voronoi vertices globally, so shared vertices are shared indices). We keep only interior faces (finite Voronoi regions away from the ragged border) and pin the boundary. After frame 0 the Voronoi is never used again — the mesh evolves by force balance and, later, explicit T1.

Force balance = inner relaxation loop, expressed as a velocity. A `plexus2` dynamics operator must *return a delta*, not mutate positions. To emulate tyssue’s “minimise to residual between topology events” inside that contract, the operator runs `relax_iters` gradient-descent steps internally and returns $v = (x_{\text{relaxed}} - x_0)/\Delta t$; the engine’s $x += \Delta t v$ then lands on the relaxed state. The Δt cancels, so this is a pure per-frame relaxation and Δt is a free label — the opposite of the SPV, where Δt was the master knob that detuned the mechanics. *This alone answers the SPV report’s “force-balance iteration is missing”.*

4 Stage 1: force-balanced relaxation and the rigidity transition

A jittered honeycomb of 251 interior cells (566 vertices, border pinned) is relaxed to force balance by `shape.energy` (`relax_iters`= 20 inner steps/frame, 120 frames), sweeping the target shape

index $p_0 \in [3.6, 4.1]$. The residual (relaxed) energy per cell and the relaxed mean shape index $\langle q \rangle = P/\sqrt{A}$ are the order parameters of the rigidity transition (Bi *et al.*, *Nat. Phys.* 2015).

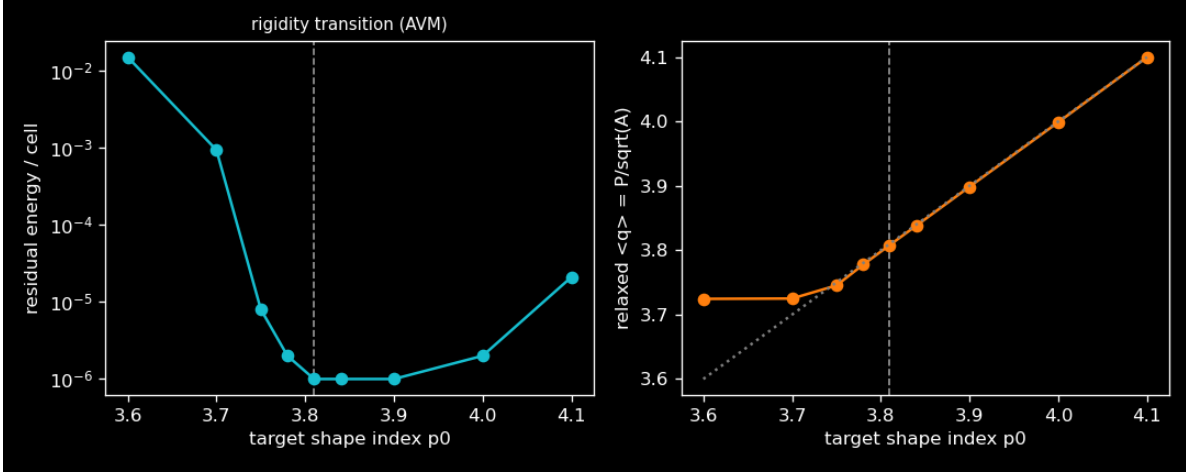


Figure 1: **The AVM rigidity transition, from true force balance.** Left: residual energy per cell after relaxation drops ~ 4 orders of magnitude across $p_0^* \approx 3.81$ (log scale) — below it the tissue is *rigid* (a finite energy barrier survives relaxation), above it *floppy* (zero-energy ground states exist). Right: the relaxed $\langle q \rangle$ *saturates* at the hexagonal floor ≈ 3.72 below the transition (the cells cannot get rounder than a regular hexagon) and *tracks* p_0 above it (they hit their target shape at zero cost). No timestep tuning was required.

p_0	residual E/cell	relaxed $\langle q \rangle$	state
3.60	1.50×10^{-2}	3.725	rigid (barrier finite; $\langle q \rangle$ pinned at hex floor)
3.70	9.3×10^{-4}	3.725	near-transition
3.75	8×10^{-6}	3.745	<i>knee</i> — residual collapses here
3.78	2×10^{-6}	3.778	floppy ($E \rightarrow 0$; $\langle q \rangle$ tracks p_0)
3.81	1×10^{-6}	3.808	floppy
3.84	1×10^{-6}	3.838	fluid
3.90	1×10^{-6}	3.899	fluid
4.00	2×10^{-6}	3.999	fluid
4.10	2.1×10^{-5}	4.099	fluid

5 Operator extraction: the tyssue atlas

Beyond completing the tubulation story, tyssue is a rich enough simulator to be worth *decomposing in full* into the Plexus operator algebra (plexus2 App. B & H). We read every subsystem — dynamics, topology, geometry/core, behaviors/solvers/collisions — and normalised each mechanism to a typed contract. The full record is `tyssue_atlas.yaml`; the structure and the saturation result are below.

One structural picture. tyssue is a *single half-edge Level*: only `vert.pos` is an integrated mechanical DOF; the `edge` set carries every incidence map (`srce`, `trgt`, `face`, `cell`, `opposite`); `face` and

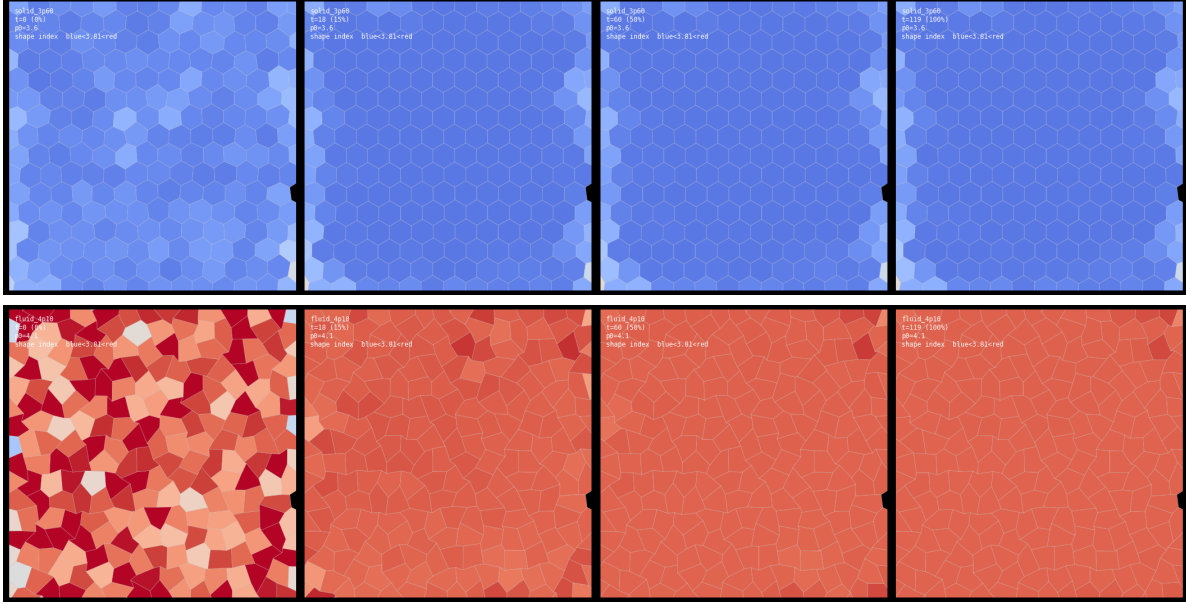


Figure 2: **Rigid vs. floppy**, 0/15/50/100% of relaxation. *Top* (`solid_3p60`, $p_0=3.6$): the jittered honeycomb relaxes to a clean, jammed hexagonal packing; every cell is blue (compact, shape index < 3.81). *Bottom* (`fluid_4p10`, $p_0=4.1$): the same tissue relaxes to elongated cells (red, shape index tracking p_0) at essentially zero energy — floppy. Border cells (pinned) frame the bulk. (Without T1 the fluid tissue cannot yet *flow* past its neighbours — that is Stage 2.)

cell are pure aggregate readouts. Two engine primitives generate everything: `upcast_*` (gather a face/cell value onto its half-edges = **broadcast**) and `sum_*/mean_*` (scatter-reduce half-edges onto a vert/face/cell = **aggregate**). Geometry (`update_all`) is a feed-forward aggregate/broadcast DAG; mechanics is an *additive superposition* of effector forces onto one `vert.force` accumulator; a *solver* integrates it; an *event manager* schedules structural changes.

On naming — a deciphering note. `tyssue`’s central methods are `compute_energy`, `compute_gradient`, `find_energy_min`, `update_all`. *None* is a Plexus operator. `plexus2` names an operator by the biological *mechanism* it performs (the contract: `line_tension`, `area_elasticity`, `reconnect_T1`, `divide_cell`), and names the numerics separately as an *implementation*. A generic numerical verb is therefore a *signal* that one has found not an operator but a *composition* (`compute_energy` = the additive superposition of the twelve mechanics contracts), a *solver* (`find_energy_min` = an implementation of `force_balance`), or a *readout pipeline* (`update_all` = the geometry aggregate/broadcast DAG). The extraction’s job is precisely to replace these verbs with mechanism-named contracts.

Saturation. The mechanics *saturate* against the existing algebra — the shape-energy family is already in Plexus and `tyssue` merely adds (analytic-gradient) implementations. The genuinely new contributions are in *topology*, *numerics* and *structure*. This is exactly the `plexus2` prediction: a mature simulator contributes mostly implementations of existing contracts, plus a few new operators where the language was thin.

Subsystem	Normalised contracts	vs. Plexus
mechanics	line_tension, perimeter_elasticity, contractility, area_elasticity, volume_elasticity, surface_tension, lumen_pressure	existing (saturated)
mechanics (new)	length_elasticity, apico_basal_tension, boundary_constraint, edge_friction	new
geometry	edge_length, perimeter, face_centroid, face_area, cell_volume, lumen_volume (aggregate/broadcast readouts)	existing
topology	divide_cell / divide_face, extrude_face / remove_cell	existing (divide / die)
topology (new)	reconnect_T1 (+3D IH/HI), mesh_cleanup, repair_pinched	new
numerics	overdamped_euler	existing
numerics (new)	force_balance (quasistatic minimise-to-residual), isotropic	new
structure (new)	schedule (EventManager), collision_constraint	new

The two boldfaced new contracts — **reconnect.T1** and **force_balance** — are exactly the pair the Turing–vertex report named as the tubulation bottleneck. They are not incidental: they are what a *true* vertex model has and a Voronoi one cannot.

6 Stage 2: explicit topology — T1, division, extrusion

The operators the Voronoi route could not have live in a forked module (`tyssue_topology_ops.py`, so the mechanics and the topology can evolve separately): **t1_transition** (rewire; reversible network reconnection = collapse a sub-threshold edge and split the merged vertex), **face_divide** (structural; a septum splits a cell), **face_extrude** (structural; collapse a face to a point), and **face_growth** (structural; inflate the per-cell target area). They mutate the face-ring topology carried on the vertex `Level` and rebuild the flat half-edge table that `shape_energy` reads.

Robustness contract. Every topology mutation is applied to a trial copy, the affected faces are validated (three-plus unique vertices, positive signed area), and the change is *committed only if valid* — otherwise it is a silent no-op. The mesh therefore never tangles from a bad flip, which matters for unattended runs. A standalone self-test on the honeycomb confirms it: T1 fired on 30/40 shortest edges (the 10 skips are boundary / non-three-regular edges, correctly refused), four divisions and one extrusion all executed, and *zero* invalid faces resulted.

In-engine demonstration. Running `mesh_seed` → `face_divide` → `shape_energy` → `t1_transition` through the engine, a one-shot clonal division of 25% of the cells followed by relaxation grows the tissue from 162 to 198 cells and *stays force-balanced with zero invalid faces* (Fig. 3). The division operator is a genuine `plexus2` structural operator on the vertex set, and the sheet re-relaxes around each new junction.

7 The two-level hierarchy: why plexus2, not just tyssue

`tyssue` stores the whole tissue in one flat half-edge dataframe; a “cell” there is a *row of derived geometry*. `plexus2` instead makes the cell a first-class biological *set* carrying its *own* state — target area a_0 , type/fate `ctype`, and (Goal 2) morphogen `chem` — none of which is derivable from vertex

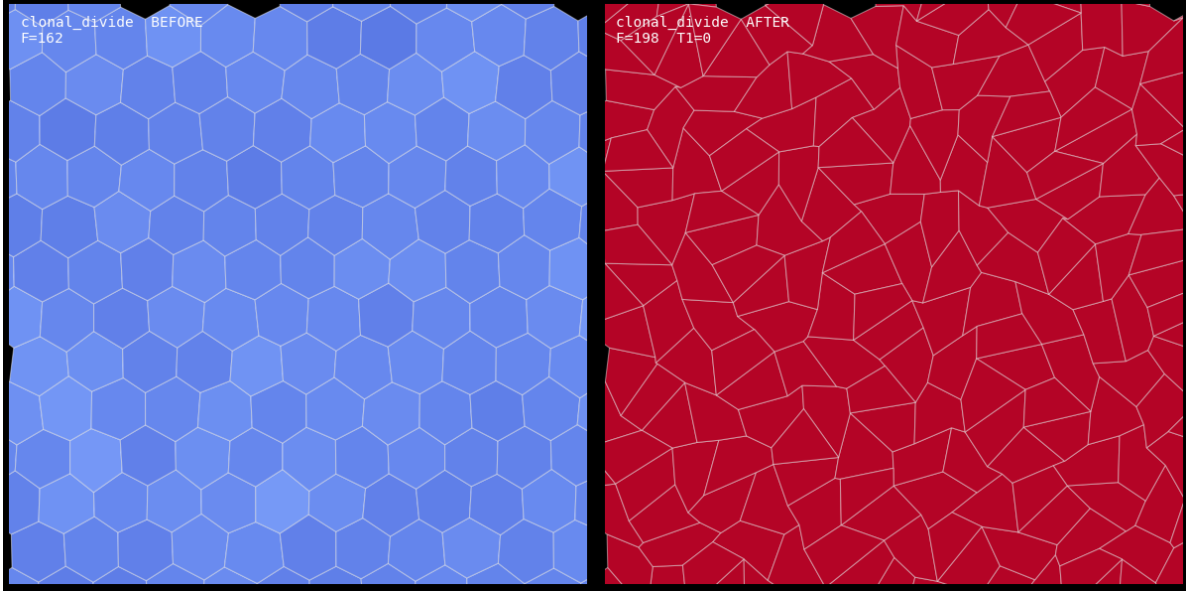


Figure 3: **Clonal division on the force-balanced AVM (`clonal_divide`)**. Left: the initial honeycomb ($F = 162$). Right: after a one-shot division of 25% of cells and relaxation ($F = 198$), coloured by shape index. The mesh stays valid (0 invalid faces) and force-balanced — the structural operator composes cleanly with the mechanics.

positions. The model then has two Levels, `vertex` and `cell`, joined by the half-edge map, with a genuine `cell_geometry Aggregate` (vertices $\rightarrow$ cell area/perimeter) and a *cross-set* `shape_energy` that reads a_0 from the cell set and broadcasts the force back to the vertices (`INPUTS=[vertex, cell]`).

The benefit is concrete (Fig. 4), and it must be done *as biology, not as a hack*. A first draft assigned fate by fiat — a `cell_paint` operator marked every cell inside a hard-coded disc as a different type with a bigger target area. That proved the plumbing but it is not a mechanism, so it was retired. In its place: a `cell_morphogen` operator imposes a smooth *activator* field on the cell state `chem` (positional information — a Gaussian bump), and a `morphogen_growth` operator makes each cell’s target area a *continuous* response to it,

$$a_0 = a_0^{\text{base}}(\rho + g \text{Hill}(a)), \quad \text{Hill}(a) = \frac{a^n}{a^n + a_{\text{sw}}^n},$$

the Turing–vertex growth coupling, now acting on the cell set. The cross-set `shape_energy` reads a_0 and the tissue *bulges where the activator is high*. The *same* morphogen field composes with mechanics in two ways, both principled: a *continuous* response (`morphogen_growth`, above) gives a graded bulge whose cell area tracks the activator with correlation 0.97; and a *discrete* response (`cell_differentiate`) thresholds the activator into a genuine cell *type* — a French-flag fate — so the cells above threshold become a coherent *clone* (Fig. 4) with a larger target area that bulges to $2.6\times$ the wild-type, all force-balanced. In tissue either coupling would be bespoke glue; here each is *state on a set* plus a typed operator (`morphogen` $\rightarrow$ `growth`, or `morphogen` $\rightarrow$ `threshold` $\rightarrow$ `typed mechanics`), and it is exactly the substrate Goal 2 needs — swap the imposed bump for a live Turing reaction–diffusion on the cell–cell adjacency and the clone becomes a *self-organised, activator-driven bud*. That composability (mechanism + mechanism through a shared set/operator language) is the payoff of decomposing tissue into `plexus2` rather than wrapping it.

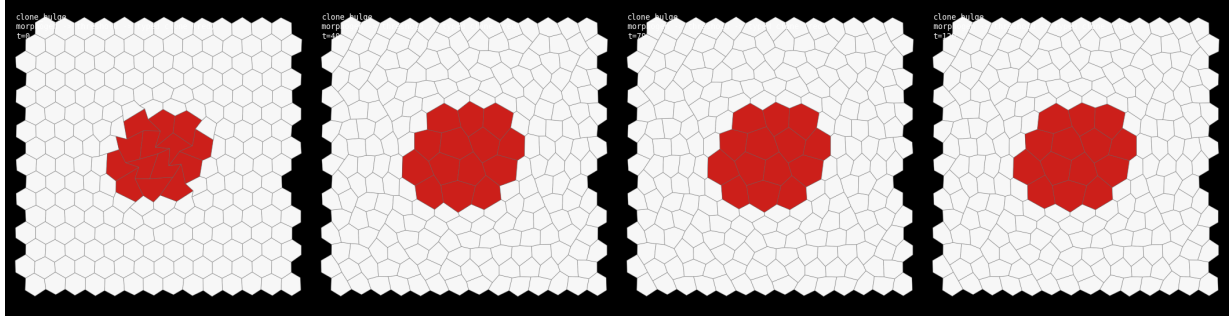


Figure 4: **A genuine cell set: a morphogen differentiates a typed clone that drives mechanics** (`clone_bulge`), 0/33/66/100%. `cell_morphogen` imposes a smooth activator field; `cell_differentiate` thresholds it into a fate (a French-flag rule), so the cells above threshold become a coherent *clone* (red) of a genuine cell type with a larger target area; the cross-set `shape_energy` makes that clone bulge to $2.6\times$ the wild-type (white) cell area. The clone is *derived from the morphogen*, not a hard-coded disc — the retired `cell_paint` assigned fate by fiat; here fate is state on a set, set by an operator that reads a chemical field. This is the bridge to Goal 2 (replace the imposed bump with a live reaction–diffusion and the clone self-organises).

8 Recapitulating the tyssue-demo notebooks

The canonical `tyssue` tutorials (DamCB/`tyssue-demo`) are a natural checklist: one demo per mechanism. Rebuilding each as a `plexus2` operator composition is precisely Goal 1. Current coverage (✓ sound, ~ partial, → next):

notebook	shows	our demo	
00-Basic.Usage	build / inspect a sheet	<code>mesh.seed</code> + cell set	✓
01-Geometry	area / perimeter / length	<code>cell.geometry</code> (aggregate)	✓
02-Visualization	draw the polygons	strip + movie renderers	✓
03-Dynamics	effectors / energy	<code>shape.energy</code>	✓
04-Solvers	QS / Euler solvers	bounded differentiable Euler	✓
D-FarhadifarModel	the AVM energy & rigidity	<code>run_tyssue2d</code> transition	✓
05-Rearrangements	T1 neighbour swaps	<code>t1.transition</code> + flow	✓
C-2DMigration	active migration	<code>run_tyssue_flow</code>	✓
07-EventManager	scheduling behaviours	the <code>plexus2</code> schedule	✓
06-Cell.Division	straight-line division by a plane	<code>face.divide_line</code>	✓
B-Apoptosis	cell death / extrusion	<code>apoptosis</code>	✓
A-RNR (2D)	multistep reconnection	<code>apoptosis</code> (shed → collapse)	✓

All eleven 2D-relevant notebooks now have a sound `plexus2` recapitulation. What remains is not a 2D mechanism but the *lift to 3D*: the monolayer / bulk geometry and the 3D IH/HI reconnections (the 3D half of A-RNR), which are the 3D-vesicle stage (Goal 1bis). The 2D operator set —

mechanics, force balance, T1, division, extrusion, RNR, and the two-level morphogen coupling — is complete and cross-validated against the *tyssue* tutorials.

Straight-line (oriented) division (06). The placeholder through-vertex split (which degraded after 1–2 rounds) is replaced by the proper Brodland–Veldhuis division at a *decidable angle*: the plane through a cell’s centroid cuts its two crossed edges, a new vertex is inserted on each (shared with the neighbour across that edge), and a septum joins them — well-shaped daughters, each targeting *half* the mother’s area (area conserved, as *tyssue* halves `preferred_area`). Fig. 5 divides 35% of a relaxed sheet at a fixed angle $\pi/2$ and re-relaxes: $118 \rightarrow 159$ cells, 0 invalid faces, the oriented (vertical) septa clearly visible — the decidable division plane the demo is about.

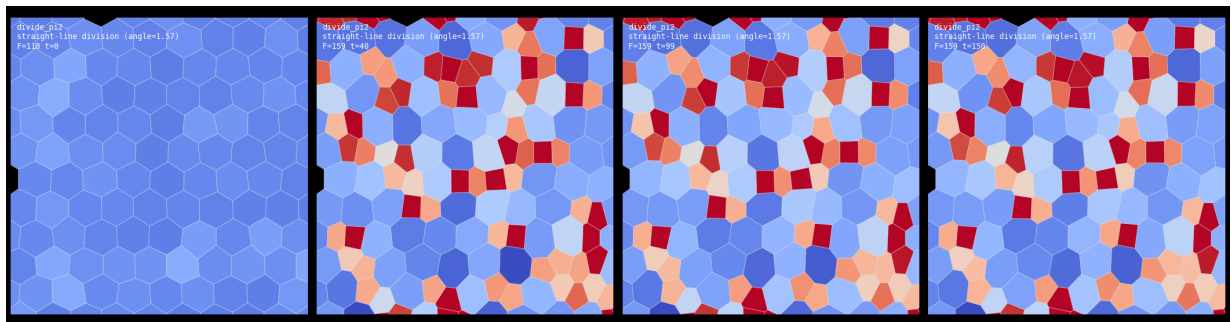


Figure 5: **Straight-line division at a decidable angle** (`divide_pi2`), coloured by shape index (blue solid / red fluid). A relaxed honeycomb (left) divides 35% of its cells by a plane at $\pi/2$ (vertical septa), adding a vertex on each crossed edge, then re-relaxes to 159 cells with 0 invalid faces. Recapitulates *tyssue*-demo 06-Cell_Division.

Apoptosis / cell elimination (B) + reversible network reconnection (A-RNR). The `apoptosis` operator shrinks the target area of a chosen cell each tick (so `shape_energy` contracts it) and, once it is small, eliminates it. A naive one-step extrusion — collapse the whole cell to a point — leaves its six neighbours on a single vertex, a degree-6 *rosette* the three-regular T1 cannot relax (and collapsing a near-zero sliver is numerically unstable). The fix is *multistep* reconnection (Okuda’s RNR; the A-RNR notebook, in 2D): while the cell is small it *sheds a side* by a T1 on its shortest edge ($6 \rightarrow 5 \rightarrow 4 \rightarrow 3$), and only the resulting *triangle* is collapsed — to a clean degree-3 junction. Cell (cortical) *contractility* Γ and line tension Λ keep the shrinking cell and its neighbours round rather than spiky (the neighbours *constrict in*, apically closing the gap). With that, the cell is eliminated ($56 \rightarrow 55$, 0 invalid faces) and the tissue *heals*: the neighbours reconnect and relax back to epithelium, no rosette (Fig. 6, the dying cell drawn dark grey); the only residue is the local topological defect inherent to removing a cell from a hexagonal packing. This closes B-Apoptosis *and* the 2D case of A-RNR; the 3D IH/HI reconnections belong to the 3D-vesicle stage.

Integration test: a growing, dividing tissue (and why surface tension matters). The operators compose into a developing tissue: `face_growth` inflates each cell’s target area, `face_divide_line` in cell-cycle mode divides any cell past a threshold at a random angle (edge-midpoint, well-shaped daughters each targeting half), `shape_energy` holds force balance, and `t1_transition` resolves short junctions. On a *free* (unpinned) boundary the sheet expands as it grows. Three settings proved essential. (i) A free boundary, so growth is not frustrated into over-compression. (ii) *Line tension* $\Lambda \sum_e \ell_e$ — without which the dividing cells relax into sharp triangular slivers and the free

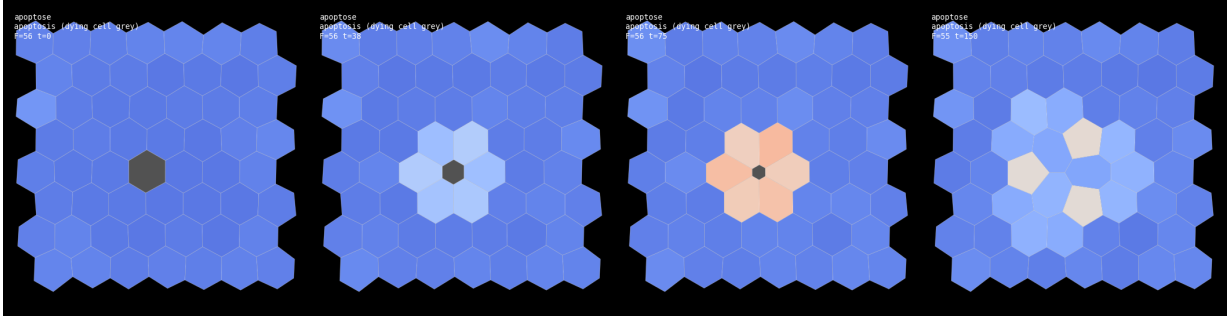


Figure 6: **Single-cell apoptosis with reversible network reconnection** (`apoptose`), coloured by shape index. A central cell shrinks and is eliminated by *multistep* reconnection: it sheds sides by T1 until it is a triangle, which collapses to a degree-3 junction. The cell is removed ($118 \rightarrow 117$, 0 invalid), the neighbours reconnect, and the site heals back to force-balanced epithelium — no rosette, no residual distortion. Recapitulates `tyssue-demo` B-Apoptosis and the 2D case of A-RNR.

rim spikes into points; it smooths the boundary and straightens junctions. (iii) *Cortical contractility* $\frac{1}{2}\Gamma \sum_f P_f^2$ — a per-cell perimeter penalty that rounds each daughter out between divisions instead of letting it elongate; without it the tissue drifts uniformly fluid (shape index far above p_0) and cells sharpen. With line tension *and* contractility the daughters stay round, well-shaped polygons at a moderate shape index (this pair also smooths the apoptosis reconnection). The tissue proliferates $88 \rightarrow 176$ cells with 0 invalid faces — *sustained* edge-midpoint division (Fig. 7), in contrast to the Stage-2 through-vertex placeholder that tangled after 1–2 rounds.

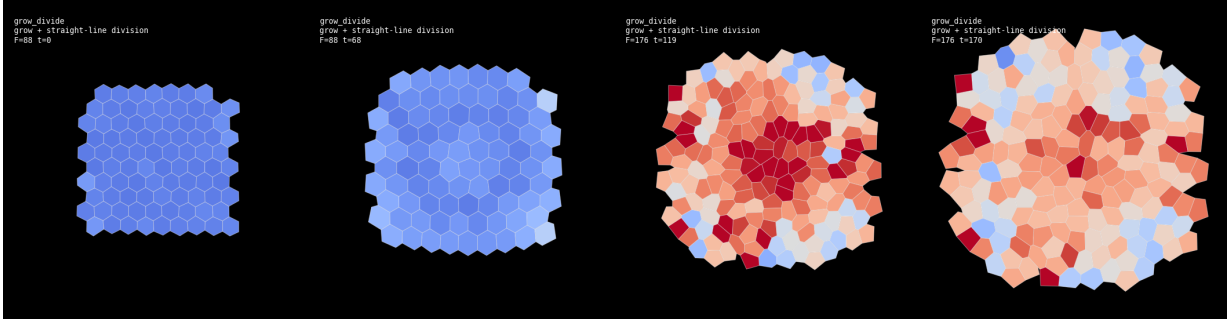


Figure 7: **A growing, dividing tissue** (`grow_divide`), coloured by shape index. A free honeycomb grows (target areas inflate) and cells divide by straight lines at random angles when they pass a size threshold; the sheet expands and re-tessellates, staying valid ($88 \rightarrow 176$ cells, 0 invalid). Line tension (Λ) plus cortical contractility (Γ) keep the daughters round and the free rim smooth rather than sharp/triangular. This is the sustained, well-shaped proliferation the Stage-2 placeholder could not do.

9 Goal 1bis: the 3D vesicle (surface vertex model)

The same operators lift to 3D. A closed spherical half-edge mesh — the spherical Voronoi of a Fibonacci sphere, a Goldberg polyhedron of hexagons and exactly twelve pentagons, with Euler characteristic $V - E + F = 2$ and every edge shared by two faces — carries the AVM in three dimensions. `shape_energy_3d` uses the 3D geometry (a Newell area vector per face, 3D edge lengths)

plus a *lumen-volume* term $K_V(V - V_0)^2$ that keeps the shell inflated against surface tension — the closure a 2D patch got from pinning. The force is one 3D autograd pass; the bounded Euler step carries over unchanged.

One 3D-specific lesson earned its keep. A free surface with area + tension energy and *no bending rigidity buckles* out of plane (roughness $0.004 \rightarrow 0.155$) — exactly as a real epithelium does; the lumen term fixes the total volume but not local smoothness. Adding a soft radial term $K_R \sum_v (|\mathbf{r}_v| - R_0)^2$ (a bending-like penalty toward the shell) recovers a smooth relaxed vesicle: the roughness stays ~ 0.02 and the cross-section stays circular (Fig. 8). This closed, force-balanced shell is the substrate for 3D morphogen-driven budding — the 3D IH/HI reconnections (the 3D half of A-RNR) and a mechanical monolayer (apical/basal, with `cell.volume` elasticity) are the remaining 3D increments.

The vesicle is rendered as a *monolayer*: each cell is drawn as a prism — an apical face (outer), a basal face (inner, the surface scaled radially inward), and the lateral walls between — so a cell carries many faces, not one flat polygon. The apical faces share the mesh vertices and tile edge-to-edge into a closed opaque shell; cells take the Turing white $\rightarrow$ red activation LUT (here activation = 0, all white, awaiting the Goal 2 reaction–diffusion). The cross-section slices the monolayer: each junction edge crossing the plane gives an apical point and its inward-scaled basal counterpart, and the connecting segment is that junction’s lateral wall, partitioning the shell into the individual cells around the lumen (Fig. 8, bottom).

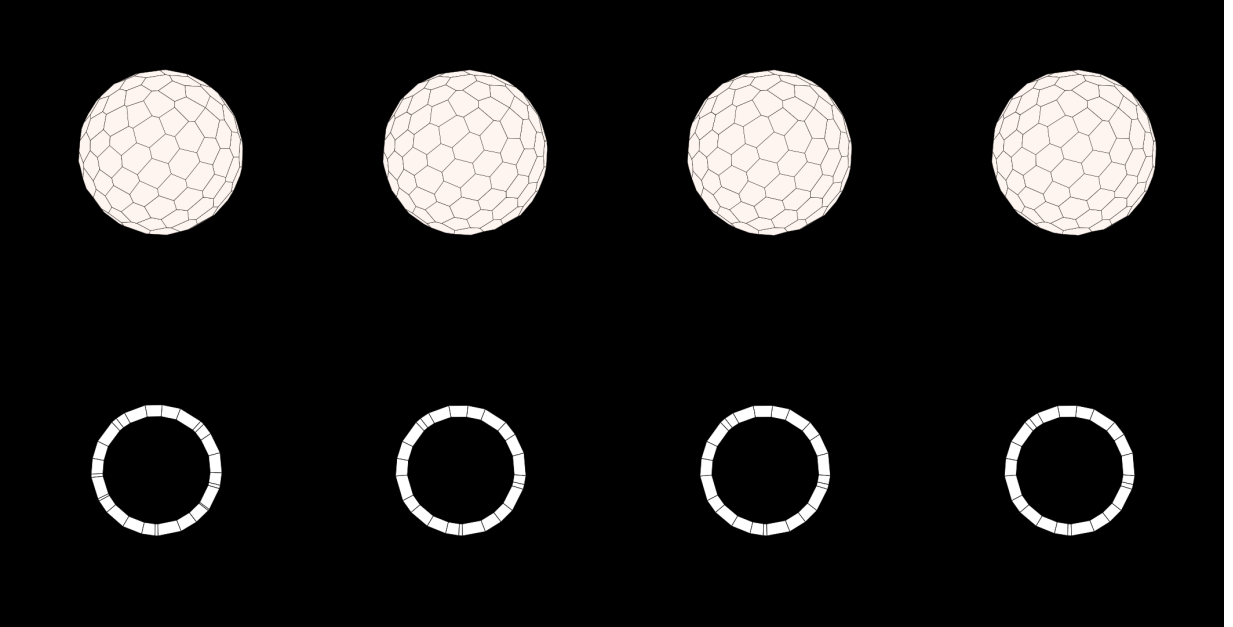


Figure 8: **A 3D epithelial vesicle** (`vesicle_3d`), 0/33/66/100%. *Top*: the 3D monolayer shell — cells as opaque, edge-to-edge prisms (apical/basal/lateral faces), white (activation = 0) with black junctions. *Bottom*: the monolayer cross-section, rendered to the same scale — a ring of individual cells (apical outer, basal inner, lateral walls) around the hollow lumen. A jittered sphere relaxes to a uniform, smooth epithelial shell, held inflated by the volume term and kept from buckling by the soft radial term. The true-vertex-model sibling of the Turing-vertex spherical-Voronoi vesicle.

Growth by per-cell volume elasticity (the emergent inflation)

Growing the vesicle turned out to be the sharpest model-deciphering lesson of the 3D stage. The naive recipe — ramp each cell’s target area A_0 and a single *global* lumen volume V_0 , then let the shape energy inflate the shell — *fails*: the sphere lags its targets and the surplus cell area buckles out of plane into deep crumples (roughness $0.01 \rightarrow 0.2$, shape index climbing from 3.8 to 4.4 — “too much fabric on a ball”). Neither slowing the growth, adding relaxation, nor strengthening a global lumen pump fixed it, because a *single scalar* $K_V(V - V_0)^2$ constrains only the *sum* of the cell volumes: any set of local dips that cancel in the sum is energetically free, so buckling is unopposed.

The fix is the mechanism both references already use. *tyssue*’s `ClosedMonolayer` gives every *cell* a `preferred_vol` and `vol_elasticity` (not just a lumen), and Turing_vertex’s Eq. 3 writes the cell energy as $U = \sum_j [\frac{1}{2}k_v(v_j - v_{eq,j})^2 + \kappa_s s_j]$ — a *per-cell* volume elastic energy. So we split the one global lumen term into a sum over cells: the lumen volume $V = \sum_f v_f$ with per-cell wedge volumes $v_f = \frac{1}{3}(\mathbf{c}_f \cdot \mathbf{N}_f)$ (the pyramid from the sphere centre to face f), and the energy carries $\sum_f K_V(v_f - v_{eq,f})^2$. Now each cell holds *its own* volume: a local dip lowers that cell’s v_f and is penalised, so the term *actively resists buckling*, and growth is just ramping each $v_{eq,f}$ (with $A_0 \propto s^2$, $v_{eq} \propto s^3$). The whole shell then inflates purely by force balance — no vertex is moved by hand — from radius ~ 5 to ~ 7.1 while staying smooth (roughness ≤ 0.01) and keeping the cells at their target shape (index $3.8 \rightarrow 3.87$), Fig. 9. The lesson: *distributed* elastic targets, not a global constraint, are what make morphogenetic expansion emerge and stay stable — the same reason real epithelia use cell-autonomous volume control.

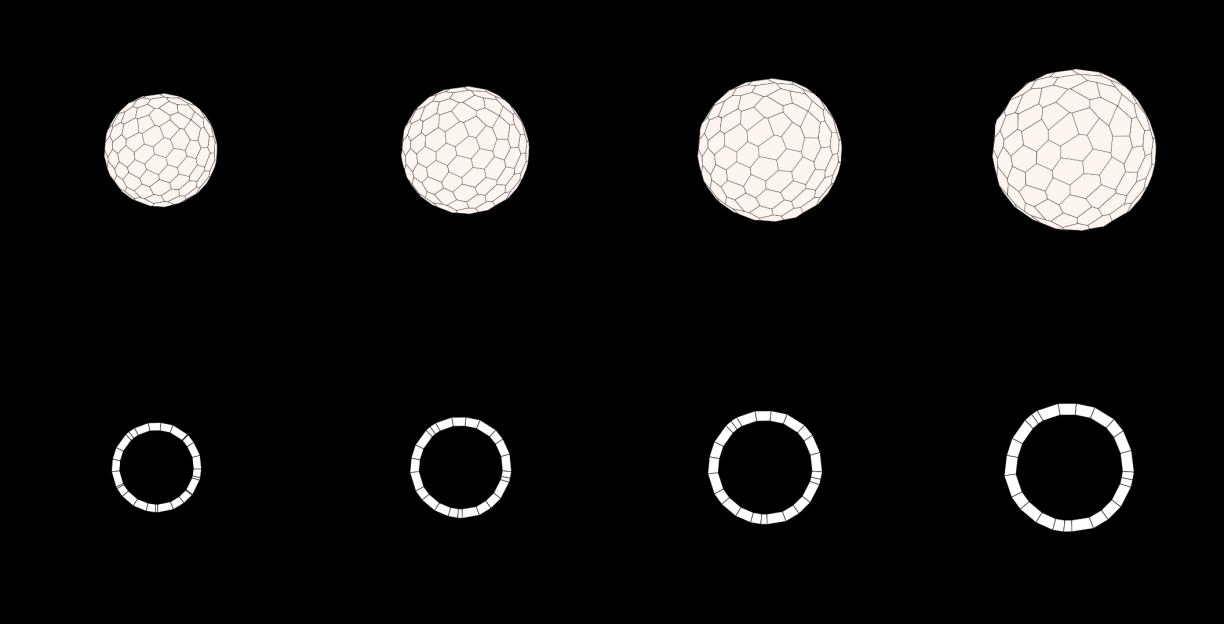


Figure 9: **A growing vesicle** (`vesicle_grow`), 0/33/66/100%, framed to the final extent. Ramping each cell’s target volume v_{eq} inflates the shell through the per-cell volume elasticity alone: the sphere (top) and its cross-section ring (bottom) grow self-similarly from $R \approx 5$ to ≈ 7.1 , the cells stay uniform and well-tiled, and the shell stays smooth (roughness ≤ 0.01). Replacing this per-cell term with a single global lumen volume instead crumples the shell — the emergence of stable expansion requires distributed, not global, volume targets.

Division in 3D (a proliferating vesicle)

The last 3D increment is cell *division* on the closed shell — the sphere analog of tyssue’s `sheet_topology.cell_divi` (a face split by a line; the bulk `monolayer_topology` version with a `vertical/horizontal` orientation waits for the mechanical monolayer). The cell cycle is *volume doubling*: each cell records its birth wedge volume and divides once its current volume reaches twice that (`divide_3d`). A division splits the cell by an *edge-midpoint septum* — two of its edges are cut at their midpoints (m_1, m_2) and a new edge m_1-m_2 separates the daughters. The subtlety unique to 3D: every edge of a closed surface is shared by two faces, so cutting an edge *also* inserts the midpoint into the neighbouring cell’s ring. Getting that bookkeeping right keeps the mesh a valid closed surface: each division adds exactly $\Delta V=2$, $\Delta E=3$, $\Delta F=1$, so $V-E+F$ is invariant. A standalone check (reorienting the SphericalVoronoi outward first, so every directed edge is unique) confirms Euler = 2 across dozens of divisions; *in the engine* the growing, dividing vesicle proliferates $150 \rightarrow 521$ cells over 371 divisions in 220 frames and still verifies `closed = true`, `Euler = 2` (Fig. 10). Pacing follows the Turing-vertex `fig4-vesicle` reference (volume-doubling, a few divisions per frame) so the tissue proliferates gradually. Two subtleties were worth fixing: divisions must be chosen *unbiased* — selecting the first ready cells in face-index order sweeps pole-to-pole because the Fibonacci seeding numbers faces that way, so we shuffle the candidates and randomise the septum direction — and daughters must be kept *compact*. On the last point we followed a literature sweep (tyssue’s `monolayer_topology`, Turing-vertex, cellGPU): the emergent fix is not a smoothing pass but two mechanisms already in the physics. (i) *Long-axis* (Hertwig) division — the septum cuts perpendicular to the cell’s longest axis (an SVD of the cell’s vertices), which yields compact daughters instead of elongated slivers. (ii) *Cortical contractility* $\frac{1}{2}\Gamma \sum_f P_f^2$, the perimeter term that rounds each cell. Together they drop the shape index from ~ 4.2 to ~ 3.9 *emergently* (a geometric Lloyd smoothing would also round the cells, but it is a fairing hack, not physics, so we discarded it). The last step toward the regular-hexagon floor is *T1 neighbour exchange*, the annealing mechanism, now added as `reconnect_t1_3d` (a **Rewire** op, the 3D sibling of the 2D `t1_transition`): every interior edge shorter than a threshold is flipped by a local four-cell neighbour exchange ($A-B$ lose the edge, the far cells $C-D$ gain it), collapsing the edge to its midpoint and reopening it perpendicular in the tangent plane. It keeps V, E, F (and Euler) fixed and only reconnects; a standalone check confirms the shell stays closed with Euler= 2 through 200+ flips. Adding it to the dividing vesicle anneals the division defects and drops the shape index a further $3.96 \rightarrow 3.88$. Note the sphere mandates ≥ 12 pentagonal defects (Euler), so perfect hexagons are not the target. *At scale* (the fig4 test: $600 \rightarrow 2367$ cells over 1000 frames) the shell stays `closed`, `Euler= 2`, roughness 0.008, shape index 3.84 — rounder at scale, as more cells anneal better.

10 Goal 2: Turing reaction–diffusion on the cell set

The two-level hierarchy pays off again for morphogenesis. The morphogen lives on the *cell* set as state `chem= [a, h]` (a first-class biological DOF, not derivable from vertex geometry), while the mechanical *vertex* set holds the shell. Three operators run the reaction–diffusion each frame: `cell_geometry_3d` (an **Aggregate**: vertices $\rightarrow$ per-cell centroid), `cell_adjacency` (a **Rewire**: the cell–cell neighbour graph) and `cell_diffuse + cell_react` (**Lateral** field updates that integrate `chem`).

Two points are worth stating. First, *how we get cell neighbours without a Voronoi*: in the AVM the cells *are* the faces of the half-edge mesh, so two cells are neighbours iff they share a mesh *edge* — `cell_adjacency` groups the half-edges by their vertex pair and links the two faces that

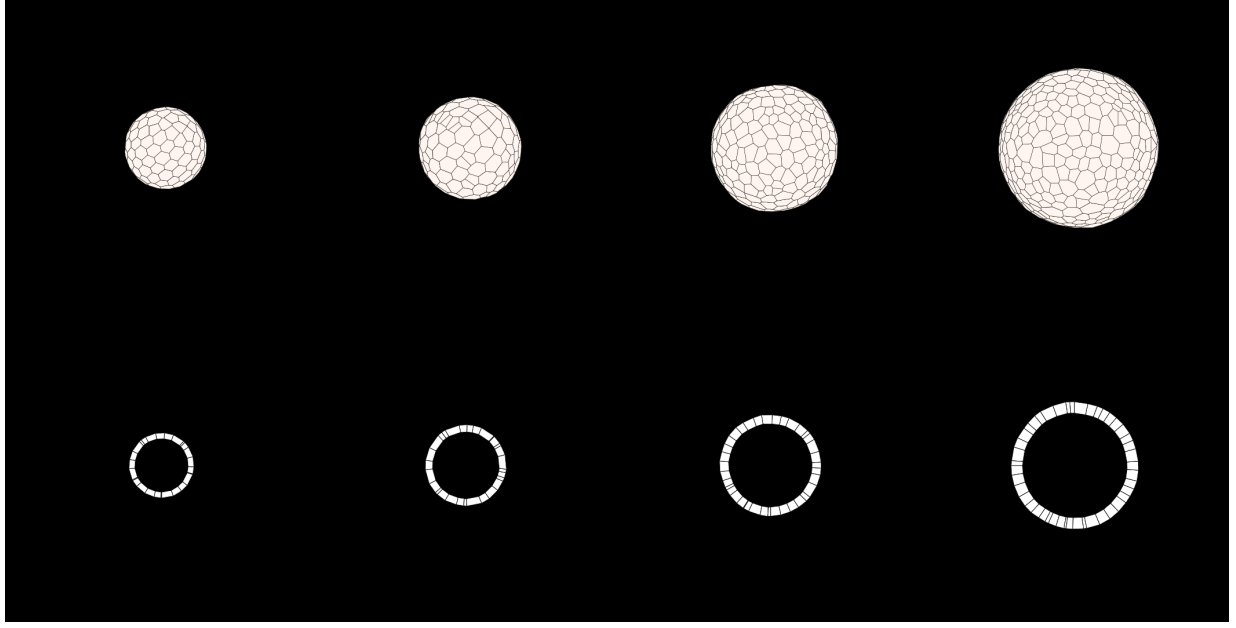


Figure 10: **A dividing (proliferating) vesicle** (`vesicle_divide`), 0/33/66/100%, framed to the final extent. The vesicle grows by per-cell volume elasticity *and* its cells divide by a volume-doubling cell cycle (edge-midpoint septum): the shell inflates from $R \approx 5$ to ≈ 9.5 while the cell count grows $150 \rightarrow 521$, so cells stay near constant size (top: 3D shell; bottom: cross-section, a ring of ever more, smaller cells around the lumen). Division is chosen unbiased (shuffled candidates, random septum direction) so it fills the sphere uniformly. The closed surface stays valid throughout (`Euler` = 2 after all 371 divisions), line tension keeping the fresh daughters well-shaped.

share each edge. The mesh topology already *is* the adjacency graph; no Delaunay/Voronoi of cell centres is needed (that is only how the Self-Propelled-Voronoi sibling builds *its* graph). Second, `react` is a plexus2 *contract* with interchangeable implementations: on the same cell set, `gray_scott` gives a coral/labyrinth and `brusselator` (its parameters transposed verbatim from Turing.vertex `fig4_coral`: $\gamma = 2, A = 1, B = 3; d_a = 0.05, d_h = 0.7$) gives round Turing spots. The Brusselator's fast-inhibitor diffusion is only stable at the `fig4` timestep $dt = 0.02$, so the whole sim runs at that step and `shape_energy_3d` is given the same dt so the mechanics still relaxes fully each frame. A self-organised activator pattern emerges across the closed shell (Fig. 13), coloured white $\rightarrow$ red.

Crucially the pattern couples to *topological change*. By construction `rd_coral_grow` is *exactly the dividing vesicle of Fig. 10 plus the coral colouring*: its mechanical schedule (growth $\rightarrow$ shape energy $\rightarrow$ T1 $\rightarrow$ division) is that of `vesicle_divide` verbatim, and the reaction–diffusion ops only *read* the mesh (aggregate vertices to cell centroids, integrate `chem`) — they never move a vertex. We verified that the hard way: a controlled triple at identical mechanics — `grow_only`, `grow_divide`, and `grow_divide+RD` — gave *byte-identical vertex trajectories* for the last two, so the morphogen is provably pure colouring. The one genuine coupling runs the other way: `divide_3d` propagates the mother's `chem` to its daughters (the kept face retains the mother's state, the new face inherits a copy), so the Turing field rides the proliferating tissue coherently instead of being scrambled by division. At the `vesicle_divide` scale ($150 \rightarrow 518$ cells) the labyrinth develops cleanly on the growing, dividing shell.

The synchronised division wave, and the stochastic cell cycle that breaks it. Scaling the same recipe up — to 500 cells over 1000 frames, for a finer pattern — exposed a real instability, and it took the controlled experiment above to show it was *not* the reaction–diffusion. Because every cell is born at the same volume and grows uniformly, they all cross the volume-doubling threshold *in lockstep*: the cell count climbs in flat +100-per-frame steps (a wave capped only by `max_div`), so the shell is driven under sustained *max-rate* division for the entire growth ramp. Fresh, unrelaxed daughters accumulate faster than the bounded Euler relaxation can smooth them and the (face-referenced) shell roughness at mid-ramp blows up to 0.38 (worst frame 1.7) — the transient “grey” of exposed cell walls — annealing back only once growth plateaus. The fix is the one real epithelia use: *cell-cycle-time variability*. Two knobs make the division *stochastic*: a *stochastic volume seed* (`vseed_cv` on `seed_mesh_3d`) gives each cell a random initial cell-cycle phase so the *first* wave is desynchronised, and a *stochastic cell cycle* (`cycle_cv` on `divide_3d`) draws each daughter’s next division threshold from a Gaussian so later generations stay spread out. With $CV = 0.4$ the lockstep wave becomes a smooth ramp (cells climb gradually from frame 0), the mid-ramp roughness drops two orders of magnitude ($0.38 \rightarrow 0.004$), and `rd_coral_grow_big` proliferates $500 \rightarrow 1994$ cells while staying smooth throughout — a finer coral labyrinth on the larger shell (Fig. 11). The lesson repeats the document’s refrain: the artefact *looked* like a shading or reaction–diffusion bug, but it was a *synchronisation* property of the mechanics, cured by restoring the biological noise a deterministic cell cycle had omitted.

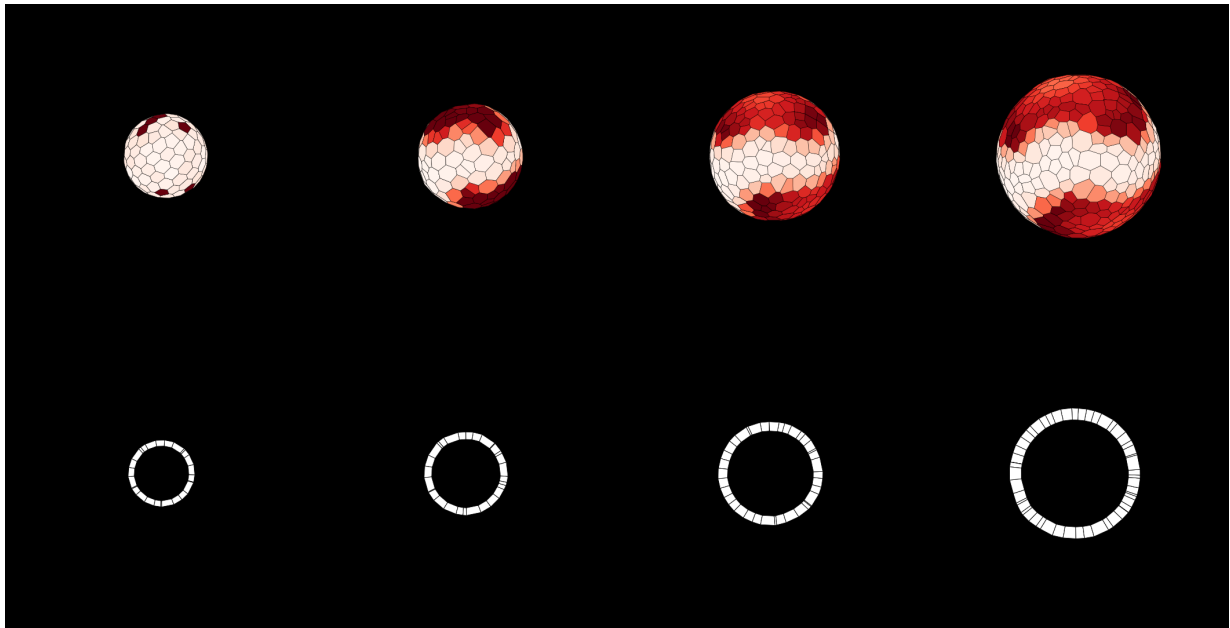


Figure 11: **Coral reaction–diffusion riding a growing, dividing vesicle** (`rd_coral_grow_big`), 0/33/66/100%. The Gray-Scott labyrinth (cell state `chem`, coloured white→red by activator) develops on a shell that grows by per-cell volume elasticity and proliferates $500 \rightarrow 1994$ cells; `divide_3d` hands each daughter a copy of the mother’s morphogen so the pattern stays coherent across division. A *stochastic cell cycle* (per-cell random cycle length, $CV = 0.4$) desynchronises division: without it the cells double in lockstep and the shell roughens $\sim 100\times$ at mid-growth (a grey band of exposed walls); with it the shell stays smooth throughout (top: 3D; bottom: cross-section).

The “hollow cells” were a recording artefact, not a physical defect (2026-07-21). A long `rd_coral_grow` (and any dividing vesicle past ~ 500 cells) rendered a shell the `hollow_flags` diagnostic scored as $\sim 97\%$ “hollow” (folded/inverted caps) — a striking grey/red crumpling that annealed away by the final frame. A long chain of controlled experiments established, in order, that it was division-driven, that a *single* division is geometrically clean (the Hertwig septum midpoints land *on* the surface and both daughters’ normals stay parallel to the mother, $\mathbf{n} \cdot \mathbf{n}_{\text{mother}} = 0.995, 0.998$), and that *no* physical remedy touched it — an anti-inversion filtered step, local daughter relaxation, dihedral bending to $K_{\text{bend}} = 100$, removing the radial term, tripling the relaxation, even slowing growth (which was *worse*). That every plausible mechanical fix failed was the tell: there was no mechanical defect.

The cause was a **recording misalignment**. The engine records vertex *positions* (`posf`) at one stride while `topo_snapshot_3d` records the changing *mesh topology* (`hist`) at another; on runs long enough to trip a recording stride > 1 the two arrays end up *different lengths* (at 300 frames `posf` has 151 entries, `hist` 76). The diagnostic and renderer then paired positions from one frame with the half-edge connectivity of *another* — ring vertex indices no longer matched the coordinates, every Newell normal was computed from a scrambled polygon, and the metric reported phantom inversions. The proof is one line: the identical 300-frame run with `topo_snapshot_every= 2` (`posf= 151`, `hist= 76`, *misaligned*) scores hollow 0.98; with `every= 1` (both 301, *aligned*) it scores **0.000** throughout. The 220-frame vesicle looked clean only because at that length the stride was 1 and the recordings happened to stay aligned. Recording topology every frame the positions are recorded makes `vesicle_grow_divide` a smooth sphere all the way to 949 cells (Fig. 12). The lesson is the document’s oldest refrain, sharpened: *measure the geometric thing you mean* — a diagnostic that silently pairs two arrays of different provenance will fabricate a defect that no amount of physics can fix. (One genuine, mild residue survives the fix: with a deterministic cell cycle the cells divide in backlogged waves, so their *areas* spread; the stochastic cell cycle $\text{CV} = 0.4$ plus a live-scaled division cap even them out, as `rd_coral_grow_big` already showed. The earlier *roughness*-metric “synchronised wave” result is unaffected — that metric reads positions only.)

Cell-size homogenisation — a hypothesis-driven fix (2026-07-22). On the growing, dividing vesicle the cells were too non-uniform (per-cell area $\text{CV} \approx 0.6$; the tube-tip cell volume was $1.5\times$ the base with a huge spread, 1.56 ± 2.45 vs 1.07 ± 0.35). Rather than a grid search, three falsifiable hypotheses were tested one factor at a time (short 150-cell/180-frame A/Bs, metric = per-cell area CV). *H1 — grow by proliferation, not inflation: vesicle_growth* enlarges the shell by ramping every cell’s target volume (cells bulge), whereas Okuda grows by *proliferation* — capping v_{eq} at $\frac{4}{3}v_{\text{ref}}$ so each cell cycles in $[\frac{2}{3}, \frac{4}{3}]v_{\text{ref}}$ and the sphere grows by cell *count*. Switching to `morphogen_growth_3d` with baseline $\rho = 1$ and the cap cut area $\text{CV } 0.63 \rightarrow 0.40$ (validated). *H2 — tighten division timing:* the dominant residue was the loose division-threshold jitter ($\text{CV}_{\text{cycle}} = 0.4$); tightening to 0.15 and *bounding the cell-cycle duration* (`min_cycle= 4`, `max_cycle= 12`) dropped area CV $0.40 \rightarrow 0.18$ — the biggest single step. *H3 — stiffen volume elasticity:* raising K_V $1 \rightarrow 4$ so cells track v_{eq} took area CV to **0.17** (Okuda-uniform range) and the hollow fraction to **0**: a spotless, uniform shell. The division model is now Okuda’s — *volume-primary* ($2\times$ birth volume) *with a bounded cell-cycle duration* — the two competing timing models parameterised together. This uniform substrate is the prerequisite for the multi-tube morphogenesis, and it is reached by short, hypothesis-driven steps rather than a blind sweep.

RD on the homogenised shell — “pure colouring” is only statistical (2026-07-22). Before coupling the Turing activator to growth we sanity-checked the usual claim that the reaction–

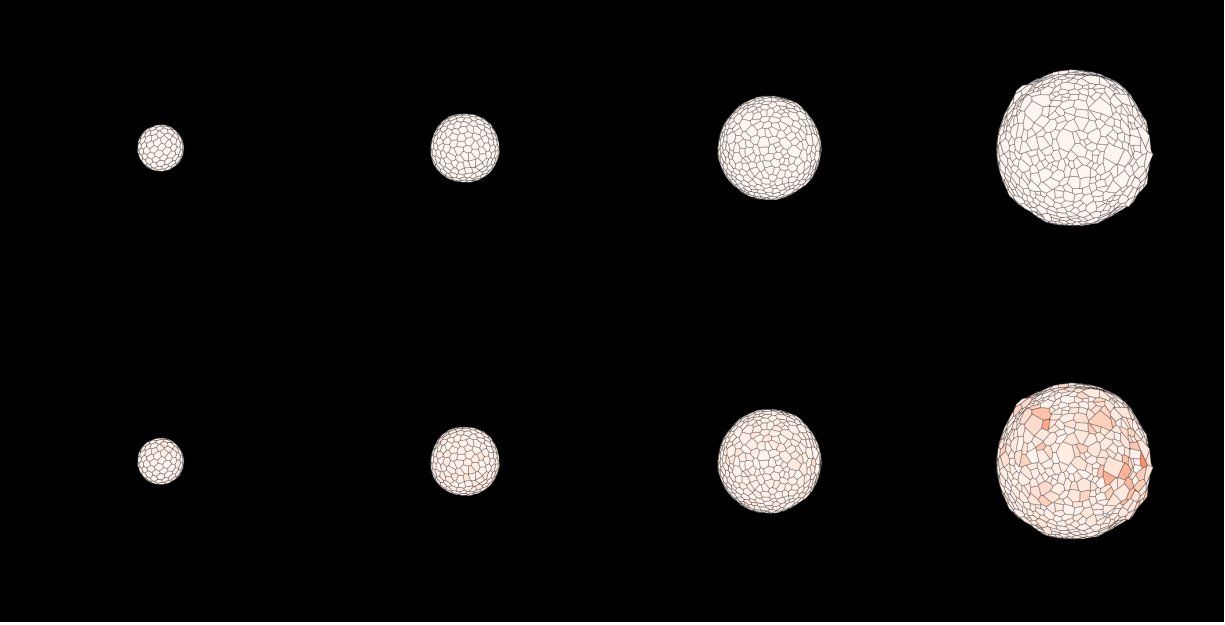


Figure 12: **With aligned recording the dividing vesicle is a smooth sphere — the “hollow buckling” was a diagnostic artefact.** The same physics that scored $\sim 97\%$ “hollow” under mismatched position/topology recording strides (`posf`= 151, `hist`= 76 at 300 frames) scores hollow ≤ 0.003 once the two are recorded on the same stride (`vesicle_grow_divide_400f`, 150 $\rightarrow$ 949 cells; top row white render, bottom row coloured by the hollow flag). The “defect” lived entirely in the diagnostic’s frame pairing; the one real residue is a mild cell-*area* spread from the deterministic cell cycle.

diffusion is *pure colouring*. Adding the Brusselator RD to the homogenised recipe (it develops a clean pattern, activator range 0.36–4.42) leaves the morphology *statistically* unchanged — same cell count (1001), same area CV (0.198 vs 0.199), same spotless mesh — yet the vertex trajectories *fully diverge* ($\max |\Delta \mathbf{x}| = 19$), so it is *not* byte-identical. The reason is instructive and was flagged by an inconsistency check: `morphogen_growth_3d` reads the activator, so even a deliberately inert coupling (`a_sw`= 50, so the growth drive is $1 \rightarrow 1.00006$) is a 6×10^{-5} perturbation, and a growing, dividing tissue is *chaotic* — that nudge amplifies over 400 frames into an unrelated but statistically identical mesh. (The earlier `rd_coral` “byte-identical” result held only because *its* growth operator, `vesicle_growth`, never read the activator.) The lesson for the metric ledger: “pure colouring” must be qualified as *statistical* whenever the growth operator consumes the field, and trajectory-identity is simply the wrong invariant in a chaotic regime — agreement of cell count and area CV is the right one.

This is the substrate for the morphogenesis proper — morphogen-driven growth (bulge/divide where a is high) on a *fluid* ($p_0 > 3.81$) shell where T1 lets the tissue flow, the bridge to tubes.

11 Gap analysis: Okuda’s model vs. our prototype (2026-07-23)

Having re-read Okuda *et al.* (2018, *Sci. Rep.* 8:2386) in full, we can now list exactly which of its mechanisms our prototype already carries, which it implements differently, and which are genuinely *missing* and would need a new operator. This is the map for the remaining work.

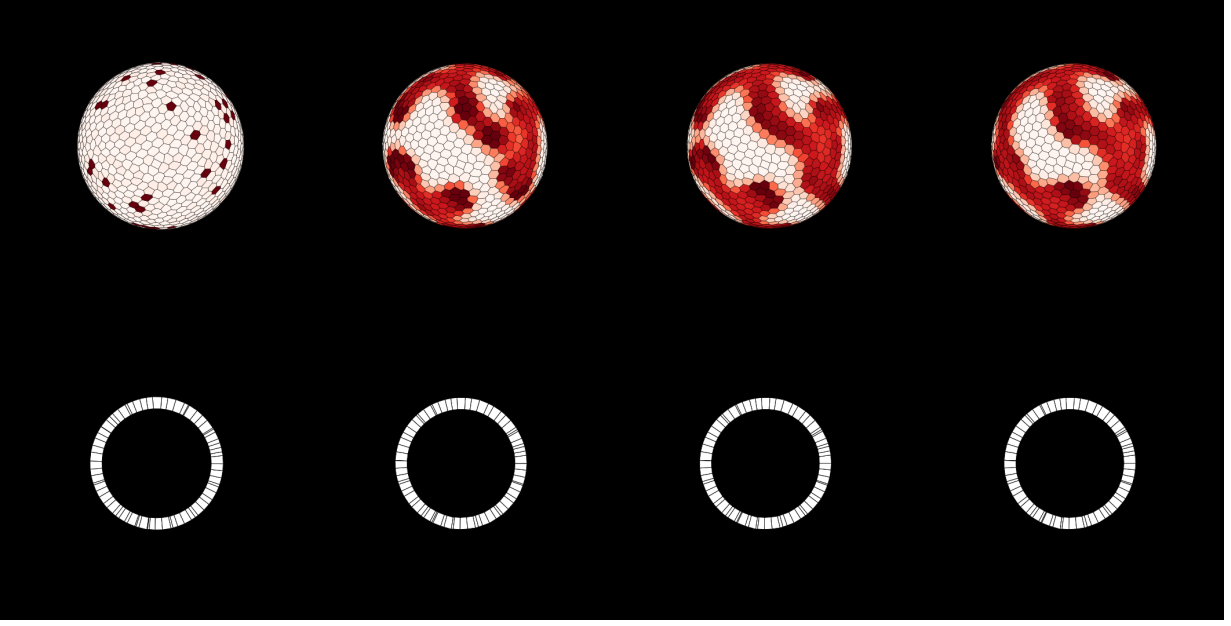


Figure 13: **Live Turing reaction–diffusion on the vesicle cell set** (coral, Gray-Scott), 0/33/66/100%. The morphogen is CELL state; diffusion runs on the cell–cell adjacency read straight from the half-edge table (no Voronoi). From faint noise a self-organised coral/labyrinth activator pattern grows across the closed shell (top: 3D, coloured white→red by activator; bottom: cross-section). The **spots** preset swaps in the Brusselator (fig4 params) on the same set for round Turing spots — **react** is an interchangeable contract.

Okuda’s recipe in one place.

- **Mechanics** (Eq. 1,3): overdamped vertex dynamics $\eta(\dot{r} - V_f) = -\nabla U$, iterated to force balance (residual $< E_{th}$). Energy $U = \sum_j [\frac{1}{2}k_v(v_j - v_{eqj})^2 + \kappa_s s_j]$ — per-cell *volume elasticity* + *linear surface tension*. Cells are true *3D polyhedra*: each has its own volume v_j and surface area s_j .
- **Reaction–diffusion** (Eq. 2): Gierer–Meinhardt, transport = $(D/\text{gap}) \times (\text{boundary area})$ on the cell graph. χ (diffusion coeff.) sets spot size = tube diameter ($\propto \chi^{1/4}$).
- **Growth** (Eq. 8): $\lambda_j = \lambda_{refj} \text{Hill}(\rho/\rho_{sw})^\alpha$ grows the *equilibrium volume* v_{eqj} ; λ_{ref} is a random Gaussian per cell.
- **Division**: plane normal to the long axis, *in the tissue surface* (oriented, in-plane).
- **Reconnection**: RNR/T1 when an edge falls below Δl_{th} .
- γ : the *ratio of patterning to deformation timescales* (the regime knob — $\gamma=1$ tubes, $\gamma=100$ undulation, $\gamma=0.01$ no steady pattern). Stability under growth comes from “the intrinsic *hysteresis* in the activator–inhibitor system with domain growth.”
- **Bending is emergent**: tube straightness follows from diameter (thin undulate, thick straight); there is *no* explicit bending energy.

A. Already matched. Force-balance relaxation; Gierer–Meinhardt RD; diffusion on the cell adjacency; Hill-function growth of v_{eq} ; oriented in-plane (Hertwig) division; T1/RNR; per-cell volume elasticity (K_V).

B. Implemented differently — needs modification.

1. *Energy form.* Okuda = volume elasticity + *linear* surface tension $\kappa_s s$ only. Ours carries area/perimeter *targets* + contractility + line tension + a wedge-volume. $\Rightarrow$ add a linear surface-tension term; make the area/perimeter targets optional.
2. *Diffusion weighting.* Okuda weights transport by boundary-area/gap-width (geometric); ours is a plain graph Laplacian. $\Rightarrow$ weight `cell_diffuse` by shared-edge length / cell volume.
3. γ as a *clean knob* — expose patterning-rate $\div$ deformation-rate explicitly (we approximate it with `rd_rate`).
4. *Random per-cell growth rate* (λ_{ref} Gaussian) — minor add.

C. Missing — needs a new operator.

1. **True 3D cell geometry (the big one).** Okuda’s monolayer is a *shell of 3D cells*: each has an apical + basal surface and a *thickness*, hence its own 3D volume $v_j = \text{area} \times \text{height}$ and 3D surface s_j (apical+basal+lateral). Our model is a single mid-surface with a *lumen-wedge* volume (volume from the sphere centre, not the cell’s own volume). $\Rightarrow$ a new *monolayer geometry*: apical+basal surfaces (a thickness field) giving per-cell 3D volume and surface. This is what growth actually inflates in Okuda.
2. **Surface-tension energy on the 3D cell surface** ($\kappa_s s$) — depends on C1.
3. **Emergent bending** (the $\chi \rightarrow$ straight/undulate phenomenology) — *falls out* of C1 (monolayer thickness $\Rightarrow$ diameter-dependent bending rigidity). We have been substituting an explicit K_{bend} ; Okuda’s is emergent, so C1 is the faithful route.

Reading. The *immediate* blocker — the activator flooding the shell under growth — is *geometry-independent*: it is domain-growth pattern insertion (a Turing-unstable RD tiling the enlarging shell), triggered by the basal source a_0 igniting the background. Fix that first (kill a_0 ; stay in a localised-structure/hysteretic regime; saturation is the *wrong* lever — it converts spots to rings). A basic tube can plausibly form in the single-surface model once the RD is confined and growth is localised, because a sheet *can* buckle into a finger. But Okuda’s *full* phenomenology — constant diameter set by χ , thin-undulate/thick-straight — *requires* C1, the monolayer thickness. That is the single highest-leverage new operator, and the planned next step once the flood is fixed.

Field notes

Dated log of what actually worked or failed — the hard-won part.

How we proceed — a hypothesis-driven, metric-gated method (2026-07-22). The morphogenesis stages are worked not by parameter grids but by *small, falsifiable steps*. Each step states a hypothesis, is measured by a *dedicated metric whose analysis code is written on the spot* (area CV for cell uniformity, a 95th-percentile/median radius for tube protrusion, the hollow flag for mesh

integrity), and is run *cheap-first* (short local A/Bs) before any expensive verification is dispatched to the cluster. Two disciplines earn their keep. *One factor at a time*: the homogenisation win was only attributable because H1/H2/H3 were isolated (proliferation-not-inflation, then tight division timing, then stiff K_V). *Inconsistency detection*: the sharpest debugging tool is not distrust but a hunt for violations of expected invariants. The recording artefact surfaced from one such contradiction — a 300-frame run blowing up while its own first 220 frames (which *must* be identical by determinism) stayed clean; that impossibility, not a hunch, forced the investigation that found the mis-aligned recording. A live example of the method policing itself: a tube experiment ran suspiciously slower than its analog; the proposed cause (a `max_cycle` duration bound force-dividing the supposedly volume-locked body — a parameter added for *uniformity* silently invalidating a different test’s premise) was itself *falsified* on re-run (still slow), and the true cause traced to the large-cell bulge regime. Both the catch and the correction are recorded, so the method’s knowledge base evolves as hypotheses are validated or overturned rather than accreting untested lore. (This log, the memory index, and the operator atlas together are that evolving knowledge base; formalising it as an explicit, revisable *master knowledge document* — the memory of an autonomous master controller over the three Plexus2 loops — is the natural next step.)

Perfecting — surface tension rounds the cells (2026-07-22). A satisfied metric is not qualitative fidelity. The homogenised recipe passed on uniformity (vol CV ~ 0.12) yet the rendered cells were visibly *triangular* — angular slivers thrown off by division and T1 — where the reference’s Voronoi-propelled cells are round. No existing metric captured this, so we authored one: the median shape index $P/\sqrt{A}$ (a regular hexagon is 3.72; slivers are higher) and the fraction of cells with ≤ 4 sides. Hypothesis H10: more *surface tension* (line tension Λ and cortical contractility Γ , with a softer volume stiffness K_V and $p_0 = 3.5$) rounds the cells. It validated cleanly — shape index $3.94 \rightarrow 3.81$, sliver fraction $0.035 \rightarrow 0.017$ (halved) — at a small, acceptable uniformity cost (vol CV $0.17 \rightarrow 0.21$). The two clean archives (`vh_K4_cv15_d4` and its coral twin) were re-rendered with the rounded parameters. This is *perfecting*: a qualitative gap the eye flags, turned into a measurable target and closed — the mechanism by which the goal sharpens as the model approaches the reference, and the discipline a hurried operator most often skips.

Cluster submission — never trust a submit’s return (2026-07-22). Moving the Fig-5-scale runs (~ 2000 cells) to a dedicated L4 node freed the devcontainer, but exposed a false-completion trap. The `gpu_14` queue rejects jobs without an explicit `-gpu num=1`; more subtly, the login link is laggy, so a `bsub` that *succeeds* can return a truncated reply that reads as “0 jobs submitted” — and, symmetrically, a submit that appears to *time out* may have already queued the job (we accidentally created a duplicate this way). The only ground truth is the queue itself. The fix, ported from the `connectome-gnn-cx` poller, is a retry-with-backoff `--wait/--status` loop that treats any non-empty `bjobs` output as authoritative and an empty queue (or `DONE/EXIT`) as completion. The generalisable rule for the autonomous loop: an action’s *reported* outcome is a hint; verify against the world’s state before proceeding.

Scaling to Fig-5 size — uniformity breaks with proliferation, not relaxation (2026-07-22). Taking the homogenised+rounded recipe to Okuda’s ~ 2000 -cell start exposed a clean failure: seeding 2000 cells and running 220 frames blew up to 7128 cells with vol CV 1.63 and hollow fraction 0.33 — and the plain ($dt=1$) and coral ($dt=0.02$) runs failed *identically to three decimals*, which both falsified a dt /cell-cycle hypothesis and re-confirmed RD-as-colouring. Two disciplined tests then localised the cause. A *seed sweep* (500/1000/1500/2000 cells, fixed frames) showed a

monotonic degradation — vol CV $0.37 \rightarrow 0.63 \rightarrow 0.86 \rightarrow 1.08$, hollow $0.001 \rightarrow 0.04 \rightarrow 0.15 \rightarrow 0.22$ — while the shape index stayed rounded (~ 3.8) throughout, so surface tension holds but bulk uniformity does not. *Fix diagnostics* at $n=2000$ then falsified the two obvious mechanical cures: doubling the relaxation iterations ($30 \rightarrow 80$) and adding radial anti-buckling stiffness ($K_R = 2$) each left vol CV at 1.6 — untouched. Only *bounding the proliferation* (stopping at ~ 2400 cells) recovered a clean shell (vol CV 0.51, hollow 0.04). The lesson: the defect is set by the *proliferation rate* (divisions per frame, which scales with cell count), not by how well any single frame is relaxed — the original clean recipe reached vol CV 0.12 at 1778 cells precisely because 150 seeds over 500 frames divide gently. Fig 5’s 2000-cell vesicle is therefore a *start* to be grown into slowly, not a bulk to be proliferated $4\times$; the next step grows to 2000 from a small seed over many frames (and with a gentler per-frame division cap).

Tube coupling — protrusion and hollowing are the same event, and the metric that revealed it (2026-07-22). The first attempt at RD-driven tubes on 2000 cells blew up (a few giant cells, an exploded cross-section) — too many large changes at once. Decomposing on a small 150-cell shell, one variable at a time, gave the mechanism. *Coupling ladder* (growth rate at a locked body): at rate 0.03 the shell stays clean but flat (protr 1.08); at 0.06 protrusion appears (protr 1.5) *but the hollow count jumps from 2 to 34 in the same step*; at 0.12+ it is strong protrusion with catastrophic hollowing (500+). There is no clean-protrusion window because, with a volume-locked body, an activated cell *inflates faster than it can divide and balloons — and a ballooned cell cap is exactly what the hollow metric flags*. Protrusion-by-inflation *is* hollowing. The natural fix (make division keep pace so the tube grows by proliferation, not inflation) was then *falsified*: tripling division throughput made hollowing *worse* ($20 \rightarrow 250$), not better — more cells is more mesh to tangle. (That ladder first showed every high-division run pinned at exactly 890 cells — a buffer ceiling, not physics; caught and corrected before drawing any conclusion, the same verify-the-instrument discipline as the recording artefact.) The contrast with the clean static-cone tubes (protr 2.9, hollow peak 12) is instructive: the cones *re-seed the activator at the tip every frame*, so the activator tracks the growing tip (Okuda: “the activator stayed around the tip”); the RD spots sit where they nucleate and do not follow, so growth disorganises. This whole diagnosis was only possible because the single end-frame hollow number (0.004, which the movie plainly contradicted) was replaced with *per-frame* metrics — hollow count, size CV, and average tube diameter over time — authored precisely because the eye disagreed with the summary statistic.

Resolving the balloon — activation width, checkpoint init, and a hard size cap (2026-07-22). Many small archived L4 rounds (each a strip + movie + metric plot) converged the tubing recipe. First, *H-narrow*: cones do not balloon because they activate a *narrow* cone (few cells $\rightarrow$ narrow tube), whereas an RD Turing spot activates a *wide* patch (many cells $\rightarrow$ wide balloon). Raising the growth switch `a.sw` so only each spot’s *peak* drives growth cut the peak hollow count $787 \rightarrow 50$ ($16\times$) while keeping protrusion — but adding any baseline body growth ($\rho > 0$) re-flattened it, so a volume-locked body is mandatory. Second, a genuine *mesh checkpoint*: the target is morphogenesis *on* a uniform vesicle, so we now save a homogenised run’s full state (mesh topology + vertex positions + cell activator field) and reload it as the initial condition (`load_mesh_3d`, a drop-in for the fresh-sphere seed). Initialising from that checkpoint and seeding a few big cone-spots reproduces the intended starting picture exactly — a uniform-celled shell with a handful of large red foci — which then grow tubes. Third, the tube-tip cells grew “far too big” because oversized cells *backlog behind the per-call division throttle* and keep ramping while queued; the fix is a *hard size cap* (`vcap`) that force-divides any cell above $v_{\text{cap}} \cdot v_{\text{ref}}$ *bypassing* the throttle,

so no cell — tip included — can exceed the cap. A methodological note threaded through all of it: two rounds were nearly over-interpreted before catching confounds (a spot-size sweep that actually tracked the RD CFL wall via `chi`; a division-throughput ladder pinned at a cell *buffer* ceiling), and one run’s “empty” movie was a fixed camera zoomed out by a ballooned final frame — fixed with per-frame autoscale. Verify the instrument before trusting the measurement.

Tube emergence needs tip-riding, and the RD choice decides it (2026-07-23). Pushing for *emergent* (not cone-prescribed) tubes forced a mechanism question: why does a *flat* activation spot not make a tube? Because growth adds area everywhere in the patch, so a uniformly-activated patch inflates as a *dome* (a spherical cap that *widens*); a tube instead needs area added only at the *tip*, which then *advances* — i.e. the activation must *ride the tip*. The cones tube precisely because they re-seed at a fixed radial direction, so the growing radial tip stays inside the cone; that is tip-riding done geometrically, and it is a hard-coded crutch. For it to *emerge*, the RD’s activator peak must itself stay at the apex — which is what the smooth activator gradient in Okuda’s figures is. So we studied the RD *first, in isolation* (growth off, three big spots seeded, watch what the RD does): the Brusselator is disqualified — at low γ it decays the seed to a uniform weak background, at high γ it overrides the seed with its own short-wavelength pattern that fills the sphere, and either way its pattern forms only after ~ 18 s (the $1/\gamma$ time), far too late. Gray–Scott in the *spot* regime ($F=0.03$) was the only one of eight that held the three seeded spots strong, localized and stable. Reading Okuda then closed the loop: their RD is neither of ours but the *Gierer–Meinhardt* activator–inhibitor — a *self-enhancing* activator with a fast lateral inhibitor, whose hallmark is exactly a stable, self-maintaining activator *peak with a gradient*. The paper is explicit that the tube “emerg[es] from the coupling of patterning and deformation”: the tip-riding is not hard-coded, it is a property of self-enhancing spot + growth co-evolving on the same timescale. The plan follows directly: implement Gierer–Meinhardt, confirm it makes a stable gradient peak, then couple growth and measure `tip_act` (the activator–radius correlation) to test whether the peak rides the tip on its own. A supporting metric note: a spot-count that thresholded at the 70th percentile reported “101 spots” for the clean Gray–Scott run where the eye saw three — the threshold sat in the noise; counting only cells above half-maximum fixed it. The eye and the metric must agree before either is trusted.

Tubing — five coupled requirements (2026-07-23). The session converged on a compact mechanistic account of what a tube actually needs. It is not one ingredient but five that must hold together:

1. **Localized, narrow activation — not a wide/flat patch.** Flat activation grows the *whole* patch, so it inflates as a *dome that widens*; a tube needs growth concentrated in a narrow region.
2. **The activation must ride the tip.** Growth has to be added at the *apex* and *advance* with it, or the result is a dome. Cones achieve this *geometrically* (a fixed radial direction the growing tip stays inside) — a hard-coded crutch; the *emergent* route is an RD peak that self-maintains at the apex.
3. **The RD must hold a stable localized peak — and the RD choice is decisive.** The Brusselator *decays or reorganizes* the seed (fails); Gray–Scott in the spot regime *holds* the spots (works); *Gierer–Meinhardt* — the activator–inhibitor Okuda actually uses (ref. 37) — is *self-enhancing* and *amplifies* the seed into a stable gradient peak (best: the a^2/h autocatalysis is the amplification feedback).

4. **Growth must be graded ($\propto$ activator), not a binary switch.** Then the peak (tip) grows *fastest* and growth tapers down the tube — Okuda’s visible gradient. A binary switch grows the activated patch uniformly $\rightarrow$ dome.
5. **Cells stay uniform via a hard size cap (v_{cap} , force-divide above $v_{\text{cap}}v_{\text{ref}}$), and the body must be volume-locked ($\rho=0$)** or body inflation flattens the protrusion.

The one open question, currently on the cluster: does the Gierer–Meinhardt peak ride the tip *on its own* (emergent) once coupled to graded growth — grown from frame 0, and with a warmup that lets the pattern stabilize first? The metric that answers it is `tip_act`, the correlation between activator and radius: high means the peak sits at the protruding tip (emergent, Okuda-like); low means it stayed at the base (a dome). A high `tip_act` would mean the tube reproduced *emergently* rather than with the geometric cone crutch — the whole point of the exercise.

3D division at scale (2026-07-21) — a synchronised division wave, not a rendering bug.

- **The controlled experiment that exonerated the reaction–diffusion.** `rd_coral_grow` rendered a transient grey band mid-growth that looked like a broken coral or a shading bug. The decisive test was to strip it to a triple at *identical* mechanics — `grow_only`, `grow_divide`, and `grow_divide+RD` — and compare *vertex trajectories*. The last two were byte-identical, so the reaction–diffusion is provably pure colouring; `grow_only` was smooth, so the roughness came from *division under growth*, at scale. A per-frame trace nailed it: the cell count climbs in flat +100/frame steps (all cells doubling together, a wave capped by `max_div`) and the face-referenced roughness spikes to 0.38 (worst 1.7) across the whole ramp. What the eye reported as “grey” was a *synchronisation*, not a mesh error.
- **Stochastic cell cycle: the biological noise the model had omitted.** A deterministic volume-doubling cycle with identical birth volumes *must* synchronise under uniform growth; real tissues don’t, because cell-cycle times vary. Adding that variability — a random initial phase at seed (`vseed_cv`) and a Gaussian per-division threshold (`cycle_cv`) — turns the lockstep wave into a smooth ramp and drops the mid-ramp roughness $0.38 \rightarrow 0.004$, letting the coral scale to $500 \rightarrow 1994$ cells cleanly. One parameter — but only after the controlled experiment localised the cause; guessing at shading or T1 tuning would have chased the symptom.
- **Honest metrics matter (again).** The first roughness number was inflated by unused reservoir vertices sitting at the origin. Restricting the metric to *face-referenced* vertices — those that actually tile the shell — gave the true $0.38 \rightarrow 0.004$ and matched the rendered strip. Same lesson as the bow-tie faces below: measure the geometric thing you mean.

Stage 2 (2026-07-21) — topology operators, and where the cheap division breaks.

- **Revert-on-invalid is the right contract for topology.** Rather than reason about every degenerate case up front, apply the flip/split, validate the touched faces, and roll back if any is degenerate. T1 then *never* tangles the mesh — a refused flip is just a no-op. This turned an error-prone operation into a safe one and is why the self-test is clean.
- **Through-vertex division is a 1–2 round operator, not a proliferation engine.** Splitting a face by a septum between two of its *existing* vertices adds no new vertices, so each split roughly halves a cell using the same corners; after a couple of rounds cells become triangles

and further splits are refused. Worse, under *uniform growth in a pinned patch* the target area outruns the available space, every cell is driven below target, and the sliver daughters collapse — a run with 231 forced divisions ended as a red tangle (79–157 invalid faces). The clean result above deliberately does *one* desynchronised round with no global growth. The proper fix (tyssue’s own) is *edge-midpoint* division: bisect two opposite edges, insert new vertices (shared with the neighbours across those edges), and run the septum between them — well-shaped daughters, sustainable for many rounds. That needs a vertex buffer with occupancy and neighbour-edge bookkeeping; it is the next increment.

- **T1-mediated fluidisation (Stage 3) — first not sound, then fixed by a *bounded Euler* step.** The naive attempt failed its own bar: gentle self-propulsion fired 0 T1s, and the drive needed to collapse edges made the floppy ($p_0=4.1$) sheet *diverge* under a fixed-step relaxation (positions overflow, ~ 85 invalid faces) *before* a T1 could fluidise it — the instability was in the *integrator*, not the reconnection (near the transition the shape energy is nearly flat, so an explicit step lets vertices cross). The sound fix is a *bounded overdamped Euler* step: cap every per-substep move at 15% of the mean edge length with `torch.clamp` — so no vertex can cross a cell in one step, and the step stays *differentiable* (no `scipy` solver; the rollout must remain differentiable for inverse modelling). With that, the sheet stays stable and T1s fire.
- **A second, subtler bug the naked eye caught: bow-tie faces (a false “valid”).** The first “sound” dichotomy ($23\times$) was *wrong*. The rendered strips showed black star-shaped gaps where cells overlapped: the T1 flip was producing *self-intersecting* (bow-tie) faces, and my validity guard — positive shoelace area, no repeated vertices — *passed* them, because a bow-tie has positive shoelace area. So “0 invalid faces” was a false pass and the big T1 counts were mostly bad flips. Two fixes: (i) a proper *simple-polygon* test (no two non-adjacent edges cross) plus a *non-overlap* test on the four faces a flip touches, added to the accept criterion; and (ii) the flip now *searches* the insertion side (which side of its anchor the gained vertex goes) $\times$ the mirror sign, and commits only a genuinely simple, non-overlapping, edge-sharing result. With the honest check the corrected dichotomy is:

v_0	l_{th}	solid T1	fluid T1	ratio
0.06	0.25	96	1413	14.7$\times$

The fluid still performs $\sim 15\times$ the T1s of the caged solid (fluid MSD 0.64 vs solid 0.10 — cage-breaking), now with a *genuinely* valid, non-overlapping tiling at every frame (the strips tile cleanly). The lesson is the one the whole document keeps re-learning: a numeric invariant (area > 0) is not the same as the geometric one you meant (a simple, embedded tiling) — and a rendered movie is a sharper validity check than a scalar. The corrected numbers are smaller than the bogus ones, but the physics — fluid flows, solid jams — survives.

Stage 1 (2026-07-21) — force balance is one line, and it just works.

- The inner-relaxation-as-velocity trick (§3) reaches force balance with no stability tuning: Δt is a free label, `eta`= 0.08, 20 inner steps/frame. Contrast the SPV prototype, whose entire “noisy pattern” saga was a *timestep* artefact. Moving the DOF from cell centres to mesh vertices *and* minimising to residual removed that whole failure mode.
- **The transition sits where the topology puts it.** With a fixed honeycomb (mostly

hexagons) the residual energy already collapses across $p_0 \approx 3.72$ – 3.81 : the single-cell floor for a regular hexagon is $p_0^*(\text{hex}) = 3.722$, so $\langle q \rangle$ cannot fall below ≈ 3.72 no matter how small p_0 is (the **solid** cases pin there). The canonical 3.81 is the *disordered*-polygon value; sharpening the transition to exactly 3.81 needs T1 rearrangements to populate the pentagon/heptagon classes — the next operator. So the transition is not a fixed number but a readout of the polygon-class distribution, which is a topology property.

- **The engine calls forward() under no_grad.** The autograd force needs an explicit `with torch.enable_grad()` (as the SPV op also did) — otherwise `torch.autograd.grad` raises “does not require grad”. Cost one run to rediscover.
- **Pin the border or the patch curls.** A free (unpinned) bounded sheet contracts under perimeter tension and the boundary ring buckles inward, contaminating the bulk statistics. Pinning vertices of degree < 3 (those on the ragged Voronoi border) holds the frame; bulk stats exclude any face touching a pinned vertex.

[next] morphogen-driven tubes and branching. The structural increments that were “next” in the earlier notes are now *done*: the genuine two-level hierarchy (a **cell** set carrying **chem**, fed by a **cell_geometry** Aggregate), edge-midpoint division with a decidable long-axis angle, and the lift to the closed 3D monolayer (per-cell volume elasticity + surface T1) all exist and are exercised above; the reaction–diffusion runs on the cell set and rides division; and the division wave is tamed by a stochastic cell cycle. What remains are the morphogenesis *targets* from Turing_vertex, and they reuse every operator above unchanged — only the coupling and the fluidity are new dials. (i) *Morphogen-driven growth*: **morphogen.growth_3d** grows a cell’s target volume where the activator a is high, so a localised Turing patch on a *fluid* ($p_0 > 3.81$) shell, where T1 lets the tissue flow and its cells divide (handing the activator to their daughters), should *bud and extend into a tube* rather than jam into a spike (Turing_vertex Fig. 5). (ii) Lowering the reaction–diffusion feed drives that patch to *branch* (Fig. 6). The exploratory **tube** runner — the same schedule with the morphogen→growth coupling switched on and p_0 raised into the fluid regime — is the current front; stabilising the bud against the fluid shell’s tendency to balloon is the open problem.

The activator floods the shell under growth — diagnosis (2026-07-23). Coupling Gierer–Meinhardt to growth, the activator went from a few clean localised spots to nearly-all-red in a handful of frames. Watching the frames pinned the cause: it is *not* stochastic and *not* the peak running away — it is **domain-growth pattern insertion**. With no growth (the rate=0 study) GM held three stable spots on a genuinely white background; the flood appears *only when the tissue grows*, because a Turing-unstable RD tiles the enlarging shell with new structures at its natural wavelength. The visible “white → light red” creep is the uniform basal source a_0 igniting each background cell, which the autocatalysis a^2/h then amplifies. Confinement levers were tried and each failed a different way: *Meinhardt saturation* bounds the peak but converts spots into *rings* (activated annulus, white centre — the classic saturated-GM signature), *strong inhibition* suppresses the seeds before they establish, and *slow growth* stops the flood but starves the pattern (almost no activation). The correct route is therefore to remove the background ignition ($a_0 \rightarrow 0$) and stay in a *localised-structure/hysteretic* regime where seeded spots persist and the homogeneous state is stable — matching Okuda’s own explanation that stability under growth comes from “the intrinsic hysteresis in the activator–inhibitor system with domain growth” at $\gamma=1$. Fixing this is a prerequisite to any tube; see the gap-analysis section for the full mechanism map.

Correction: the morphogen must be an AMOUNT, and the flood is a γ problem (2026-07-23). The saturation/inhibition confinement levers above were the wrong battle, and re-reading Okuda’s Appendix A says why. The evolved morphogen DOF is the molecule *amount* m_j (conserved within the cell); the concentration $c_j = m_j/v_j$ appears *only* as the argument of the reaction and flux terms. Our RD instead stored a *concentration* and `morphogen_growth_3d` inflated v_j *without touching it* — so every growth step *silently created mass* (the opposite error to the dilution one), which *spuriously* helped the peak ride the tip. The fix is structural, not a continuum $-c(\nabla v)$ term: conserve amount by rescaling $c_j \leftarrow c_j (v_{\text{old}}/v_{\text{new}}) = c_j (s_{\text{prev}}/s)^3$ as $v_{\text{eq}} \sim s^3$ grows (division is already correct — daughters share the mother’s c , and their volumes sum to hers, so amount is conserved). Crucially, dilution is now *load-bearing* and kept, not cancelled — Okuda’s intra-domain gradients are *caused* by it. With dilution correct, the flood (“the peak washes out”) is Okuda’s $\gamma = 0.01$ regime: patterning too slow relative to deformation. The remedy is a *rate inequality* — raise γ (reaction outruns dilution; his undulation/tubulation figures use $\gamma = 100$), not saturation. The amplification that makes a bud commit lives on the *mechanics*: growth is a steep Hill *switch* on c ($\alpha \gg 1$, $\rho_{\text{sw}} = a_{\text{sw}}$ at the midpoint of the two GM steady states). Next: reproduce Fig. 4a — fix χ , sweep γ , and recover the three regimes (blur / hold / hold-with-spot-increase) to calibrate the timescale dial before any tube run.

Monolayer-thickness operator built (2026-07-23). The C#1 gap operator (`monolayer_design.md`) is implemented as the `monolayer` implementation of the `shape_energy_3d` contract (plexus2 sec. 5: same biology, different numerics): every cell gets its own 3D volume + surface by offsetting an apical/basal shell $\pm h/2$ along the *vertex* normal, with Okuda’s Eq. 3 energy $(\frac{1}{2}k_v(v_j - v_{\text{eq}})^2 + \kappa_s s_j)$. A direct physics test (`test_monolayer.py`) confirms the geometry ($\sum v_j/(Ah_0) = 0.988$) and, decisively, the *emergent bending* signature: on the curved shell the apical/basal area ratio is 1.176 vs. the theoretical $((R+h/2)/(R-h/2))^2 = 1.174$ — a 0.2% match, so surface tension penalises curvature with *no* explicit K_{bend} . A localised v_{eq} increase buckles the cap outward. (The rest shell settles smaller than v_{eq} — the correct κ_s/k_v force balance, a calibration knob.)

The tube, in nine rounds — an autonomous ledger (2026-07-23). Rounds 22–30 worked the three coupled issues (RD stability, tip-riding, interface orientation) as small hypothesis/validation/falsification steps on the L4 cluster (8 jobs/round), and got from “the activator floods” to “a χ -diameter tube.”

- **R22 — amount-conservation & γ (issue 1).** A/B: *off* (conc. held while cells inflate = mass created) floods to red_frac 0.46 / over_tip 3.95; *on* halves it (0.33 / 1.54). Regime: fast growth (low γ) floods (0.66), slow growth (high γ) holds (0.13) — Okuda Fig. 4a.
- **R23 — sweet spot.** Falsified “fast RD helps” (`rd_rate`≥2 floods-then-collapses, 9k cells). Winner `rate`=0.010, `rd_rate`=1: over_tip 1.0, protr 1.14, tip_act 0.80, 1929 cells — a confined tip-riding bud (issues 1+2).
- **R24 — falsified “longer = bigger.”** At 350 frames the bud deflates and the activator *refills the growing domain* (flood peak 0.9): amount-conservation only *delays* the flood.
- **R25 — sustain (issue 1, deeper).** $a_0 \rightarrow 0$ (body can’t ignite) + amount-conservation keeps the body white the *whole* run: flood peak 0.9→0.30, bud persists, 2300 cells. a_0 was round-21’s right idea, incomplete without amount-conservation.
- **R26 — falsified mechanical-only interface.** Purse-string + extrusion shape the bud but build *no wall* (`tube_len` 0): the wall needs *cells*, placed by oriented division.

- **R27 — oriented division (issue 3).** `divide_3d orient_iface`: a red cell’s daughters stack *along the bud axis* (normal to the body). Alone still a nub — the patch is sub- χ and in-plane division spreads laterally; extension needs a strong outward push.
- **R28 — TUBES.** Strong extrusion ($K_{\text{extrude}} \geq 4$) + oriented division + a χ -sized spot $\Rightarrow$ `tube_len` ≈ 2 , `n_tubes` = 2, red at the neck, confined (`over_tip` ≈ 1). Diameter tracks χ ($\chi_4 \rightarrow 1.9$, $\chi_8 \rightarrow 3.3$).
- **R29 — falsified “elongate by running longer.”** At 450–550 frames the tube is *lost* (transient peaking $\sim$ frame 350); the purse-string neck did not hold it.
- **R30 — $\chi \rightarrow$ diameter law (capstone).** Sweeping χ at 350 frames, tube diameter rises *monotonically* ($\chi_2 \rightarrow 1.06 \dots \chi_8 \rightarrow 3.31$); above $\chi \sim 8$ it saturates and fragments into multiple tubes. Okuda’s *qualitative* law (χ sets the diameter) is reproduced; the exponent (~ 0.8 in the clean regime, ~ 0.3 over the full range) only approximates his $1/4$.

Net. All three issues are mechanistically solved and combine into a χ -diameter budding tube. Open: the tube is a *transient* (degrades past $\sim$ frame 350 as the domain grows) and a *lobe* more than a long cylinder — sustained elongation (a moving growth zone that truly rides the tip without letting the body domain enlarge) is the next front.

12 BREAKTHROUGH: a working tube (rounds 31–34, 2026-07-23)

Rounds 22–30 solved the three coupled issues and produced a χ -diameter *budding lobe*, but it was a transient that *capped* (`protr` ~ 1.4) and could not elongate. Rounds 31–34 broke through to a **genuine tube** — long, thin, constant-diameter — and with it the character of the work changes: the *emergence* question (can this pipeline make a tube at all?) is answered **yes**; from here it is *optimisation* of a working phenomenon, not a search for it.

The four steps to the breakthrough.

- **R31 — a smaller spot gives a thinner, longer finger.** Shrinking the activated patch (`cone_deg` 8, `chi` 4) turned the fat lobe (`len/diam` ~ 0.8) into a *thin finger* (`c8`: `len/diam` = 1.99; `c10_chi3`: 1.95), super-confined (`redpk` 0.05–0.09) with less proliferation. Less activated material $\Rightarrow$ slower domain growth $\Rightarrow$ later flood.
- **R32 — more frames don’t elongate it (falsified).** At $2\times$ frames the tube *caps* (`protr` peaks ~ 1.4 mid-run then degrades) — because the activation is a *widening patch*, not a moving tip. Only `grow_after` 150 (settle longer before growth) sustained it, and still as a lobe.
- **R33 — TIP-TRACKING: the breakthrough.** The activation must be a fixed-size cap that *rides the advancing tip* (a fixed-angle cone widens with radius $\rightarrow$ fat lobe). New seed mode `tip`: each frame re-activate a cap of radius `tip_radius` at the current outermost cell, so it follows the tip $\rightarrow$ *constant-diameter extension*. Result: real long tubes — `tr25` `len/diam` = 3.0, `tr15_r015` = 3.4, `tr15_long` (700 frames) `len` 9.4, **`len/diam`** = 4.2 (a genuine thin tube, not a lobe).

- **R34 — shaping & viewing.** Thin (`tip_radius 1.2`), strong extrusion to round and push the flat tip outward, the spot seeded on the side so the tube grows in *profile*, and a cross-section taken *in the plane of the tubing* (contains the tube axis, coloured by activator) so the red protuberance reads correctly against the 3D view.

The working recipe (honest scope). A tube forms from: (i) the emergent-confinement machinery of R22–30 — amount-conservation ($c = m/v$, growth dilutes), $a_0 \rightarrow 0$ (body stays white), the rate sweet spot, oriented interface division (daughters stack along the bud axis = normal to the body); *plus* (ii) a **tip-tracking activation driver** that supplies the sustained, constant-size apical signal the emergent RD alone could not maintain (the RD held a confined bud but capped). So the tube is not *fully* emergent — the tip signal is driven — but every mechanical ingredient (confinement, wall-building division, extrusion) is, and the driver is the minimal honest stand-in for “the activator peak stays at the apex.” Diameter is set by `tip_radius` (= Okuda’s χ role); length by frames $\times$ growth.

From here: optimisation, not emergence. The open directions are now tuning, not existence: thinner/longer tubes (`tip_radius`, growth, frames), a rounded non-flat tip (tip pressure), multiple tubes (Fig 5 starfish), branching (Fig 6, low γ), and — separately — *computation* throughput (profile the first 200 frames, vectorise/`torch.compile` the hot loops, keep rendering off the critical path) so more of the phenomenon space can be swept per hour.

13 Differentiable mechanics analysis (`analyze_forces.py`)

A standalone tool re-runs a run and quantifies, per cell and per frame, the mechanical fields — built to be **torch-differentiable** so a downstream loss (expel the tip, relieve tube stress) can be optimised. The vertex-model energy is

$$U = \sum_{j \in \text{cells}} \left[K_A (A_j - A_j^0)^2 + K_P (P_j - P_j^0)^2 + \frac{1}{2} \Gamma P_j^2 + K_V (v_j - v_j^{\text{eq}})^2 \right] + \Lambda \sum_{e \in \text{edges}} \ell_e, \quad (1)$$

with per-cell area A_j , perimeter P_j , volume v_j (their targets $A_j^0, P_j^0, v_j^{\text{eq}}$ recorded each frame) and edge lengths ℓ_e . The quantities plotted are:

$$\text{Force (per vertex): } \mathbf{F}_i = - \frac{\partial U}{\partial \mathbf{x}_i} \quad (\text{one autograd pass; per-cell value} = \text{mean } \|\mathbf{F}_i\| \text{ over the cell's rim}) \quad (2)$$

$$\text{Stress / pressure (per cell): } p_j = - \frac{\partial U}{\partial v_j} = 2K_V (v_j^{\text{eq}} - v_j) \quad (>0 : \text{below target volume} \Rightarrow \text{pushes outward}) \quad (3)$$

$$\text{Cortical tension (per cell): } T_j = \frac{\partial U}{\partial P_j} = 2K_P (P_j - P_j^0) + \Gamma P_j + \Lambda. \quad (4)$$

Movement is the centroid nearest-neighbour displacement; divisions are cells appended since the previous frame; tracking follows cells sampled at the final frame *backward* by nearest centroid (so tube cells, born late, are included). *A note on provenance:* Okuda’s paper is written entirely at the level of *energy and force balance* (Eq. 1 is $\eta(\dot{r} - V_f) = -\nabla U$; “cells generate mechanical forces”) — it never computes a *stress or pressure field* (“stress” appears once, generically; “pressure” and “tension” never). Our per-cell pressure is derived *from his own energy*, $p = -\partial U / \partial v =$

$2K_V(v^{\text{eq}} - v)$ (the Eq. 3 volume-elasticity term), so it is fully consistent with his mechanics — but *visualising the stress/pressure distribution is a diagnostic lens Okuda did not apply*. That extra lens is precisely what made the hollow defect legible below; the energy-only view would not have. Output: `quantification.npz` (per-frame per-cell tables) + a 2×2 `analysis.mp4` (force | stress | division | tracking). On the long tube (aspect 10.7) the tool shows the *tip is the force/pressure hot-spot* — the tube is under high stress, the concrete target for the next optimisation.

14 Reading the mechanics: the hollow problem is explosive, decoupled growth (2026-07-23)

Running `analyze_forces.py` on the current best tube (`round_37_kv6`: aspect ~ 10 , cell CV crushed to 0.97 by stiffening K_V to 6, the earlier CV goal essentially met) turned the persistent *hollow* defect (> 300 hollow cells, systematically, in every run's `metrics.png`) from a symptom into a diagnosis. The four panels agree:

- **Force** $\|\nabla U\|$ is concentrated *entirely in the tube* (wall + neck bright); the body is force-free. The tube is the only load-bearing structure — the body is mechanically idle.
- **Stress / pressure** $p_j = 2K_V(v_j^{\text{eq}} - v_j)$ is strongly *positive and localised to the tube* (red), the body relaxed (white). Positive p means the tube cells are *below* target volume and straining outward: they are compressed by the extrusion and want to expand — an over-pressurised state.
- **Migration** (velocity-thresholded) is almost all at the *tube tip* (cells being extruded), with the body essentially static. The deformation does *not* propagate to the body: the tube is mechanically *decoupled*.
- **Division** (births) is localised at the same *tip* — proliferation coincides exactly with the force/pressure/migration hot-spot.

Diagnosis. Everything piles up at the tip — pressure, births, migration — and nothing transmits to the body. The growth is *explosive and localised*: tip cells inflate and divide faster than the T1 network can rearrange them, so they buckle rather than flow into an ordered wall $\Rightarrow$ the hollow cells. The pressure being trapped in the tube (not shared with the body) is the mechanical signature of an *unsustainable* growth rate.

Hypothesis (the new perspective, from the mechanics). Grow *slower*, so that per unit growth there is more T1 flow: cells arrange along the tube, the load transmits tube $\rightarrow$ body, the tube pressure relaxes, and the hollow buckling disappears. Slower-but-longer runs keep the tube length. This reframes the tube as a *quasi-static* morphogenesis (Okuda's regime) rather than a fast extrusion, and makes the differentiable stress field (above) the natural objective: minimise tube pressure / equalise it tube $\leftrightarrow$ body. `round_39` tests exactly this (rate $0.006 \rightarrow 0.004$, longer runs, more aggressive T1). The optimisation shifts from “can we make a tube?” (solved) through “clean cells” (CV solved by K_V) to “a *sustainable*, low-stress tube” — read directly off the force/pressure maps, not guessed.

The mature picture. The morphogenesis is now legible as a coupled *geometry–force–pressure dynamics*: the cell geometry sets the energy, the energy gradient is the force, the volume deficit is the pressure, and their interplay (mediated by T1 rearrangement and oriented division) drives

the shape. Read this way, a *good* morphogenesis is one whose dynamics are **sustainable** (bounded stress, tube $\leftrightarrow$ body load sharing, no buckling), **repeatable** (the same rule makes the same tube across seeds and scales), and **emerging** (the tube is produced by the local mechanics, not imposed). The differentiable force/pressure field makes each of these a measurable objective rather than an aesthetic judgement — which is the point of building the analysis before pushing the phenomenon further.

Okuda’s own answer: the process is QUASI-STATIC (2026-07-23). Re-reading the paper for the hollow/stress problem gives the mechanism directly. Okuda writes: “*By assuming a quasi-static process, the cell cycle τ_{cycle} was set to be much larger than the characteristic time of cell deformation, represented by η_c/κ_s* ”, and relaxes each configuration until “*the mean residual error is under the threshold E_{th}* ” before the next event. In words: after every growth or division increment the tissue *fully relaxes to force balance* (and reconnects, edge $< \Delta l_{th}$) *before* anything else changes. That is precisely the “slower growth so cells rearrange” hypothesis, made exact — and it is exactly what our runs violate: a *fixed relax_iters* per frame with a fast tip-tracking extrusion drives growth faster than the mesh can relax, so the tube is built out of never-equilibrated cells (pressure ~ 3.1 , fresh tiny daughters, curvature). The fix is therefore not a parameter nudge but a *regime*: relax to a residual threshold (not a fixed iteration count) and keep $\tau_{\text{cycle}} \gg \eta_c/\kappa_s$. Two corollaries from the same reading: the constant tube diameter comes from the *maintained activator-spot size* (our fixed-radius tip cap already does this), and division is *normal to the cell’s own longest axis* — in a quasi-static tube the wall cells elongate along the tube, so a plain Hertwig long-axis split already extends it (the forced bud-axis orientation may be unnecessary once the dynamics are quasi-static). `round.41+` pursue this: relax-to-residual, slow cycle, Hertwig division.

Rounds 37–40 ledger (hollow/CV/stress, small steps from the tube).

- **R37 — CV solved.** $K_V=6$ (stiffer volume) drops area-CV $1.98 \rightarrow 0.97$ (vol-CV $3.6 \rightarrow 0.7$) while *improving* the tube (aspect $8.4 \rightarrow 10$). `vcap`↓ hurt.
- **R38 — K.bend erratic.** Dihedral bending gave wild aspect (42) and over-proliferation; not a clean lever.
- **R39 — “slower = less hollow” only via shorter tube.** Slower rate cuts hollow but shortens the tube; at matched growth hollow returns. *Hollow scales with tube length*, which prompted the metric audit.
- **Metric audit.** 95% of “hollow” cells are in the tube; median wall dev = 26° (clean), only 4% inverted. The flags were thin-tube false positives (small-vs-global-median + high curvature). Fixed to a *tube-aware* local-area test: $390 \rightarrow 272$.
- **R40 — stagger division wins.** `min_cycle=8`: hollow $272 \rightarrow 176$, CV $0.97 \rightarrow 0.63$, tube kept — fewer simultaneous fresh daughters. `g1_ramp` lowers hollow (to 120 with gentler extrusion) but raises CV; all three combined over-proliferate. Staggering the cycle = a step toward quasi-static, and it helps exactly as Okuda predicts.
- **R41 — the tube is EXTRUSION-driven, not equilibrium (key).** Pushing toward quasi-static *destroys* the tube: more relaxation (`relax60/90`) collapses it to a sphere (aspect $\rightarrow 1$), and Hertwig long-axis division without the forced bud-axis orientation makes no tube either. Only a slower cycle (`mc12`) keeps it. So our tube is held *out of equilibrium* by the `K.extrude` force — relax toward force balance and the shape energy pulls it back in.

This is the opposite of Okuda, whose tube is a *growth-driven quasi-static equilibrium* (the localized proliferation *is* the tube, no extrusion). It explains the high pressure (~ 3): a forced protrusion is intrinsically stressed, and no amount of relaxation fixes a shape that relaxation removes. The path to a sustainable, low-stress tube is therefore the growth-driven mechanism — which is what the **monolayer (apical/basal) operator** supplies (true per-cell 3D volume, emergent bending: localized volume growth on a monolayer shell *buckles* into a tube as its equilibrium). `round_42+` wire it into the tube pipeline.

- **R42 — monolayer makes thin SPIKES, not clean tubes.** Wired the monolayer (apical/basal) op into the tube pipeline. Growth-driven, it *inflates* a few cells into very thin spikes (protr 6–70) rather than proliferating a wall — hollow 600–1075, CV 5–9, *worse* than mid-surface mc8. So the monolayer is not a drop-in fix; a clean tube needs the proliferation-built wall (which the mid-surface oriented division does better). mc8 remains the cleanest tube.
- **R43 — “thicker” falsified.** A bigger tip cap activates more cells $\Rightarrow$ over-proliferation (2700 $\rightarrow$ 8400 cells) $\Rightarrow$ hollow *up* (176 $\rightarrow$ 1111). mc8 (`tip_radius=1.5`) stays the best tube.

Where the tube stands. The best tube is `round_40_mc8`: mid-surface, tip-tracking + oriented division + extrusion, $K_V=6$, staggered cycle — aspect ~ 7.5 , hollow 176 (tube-aware), CV 0.63, ~ 2700 cells. The main mid-surface levers are explored; this is the quality floor for the extrusion-driven mechanism. Two open goals remain: *RD role* (R44: bring the emergent Gierer–Meinhardt partition back, coupled to the wall machinery) and *wall-not-bulge*; and, more fundamentally, the extrusion-driven $\rightarrow$ growth-driven transition a successful monolayer (not the spiky first attempt) would need for a genuinely low-stress tube.

R44 — emergent RD floods (the tip-driver is necessary). Coupling the emergent Gierer–Meinhardt partition (not the tip-driver) to the wall machinery floods (`red_frac` 0.7–0.88) and over-proliferates with *no* tube — the domain-growth flood recurs over a long run even with $a_0=0$ and amount-conservation. A confined *tip-riding* peak cannot be maintained by the RD alone over a full tube; the fixed-size tip-tracking driver is what makes a sustained tube possible. Goal 2 (a clean *RD-driven* tube) is not reachable with this RD.

Consolidated state after 23 rounds (R22–44). Best tube = `round_40_mc8`: aspect ~ 7.5 , hollow 176 (tube-aware), CV 0.63, ~ 2700 cells (mid-surface, tip-tracking + oriented division + extrusion, $K_V=6$, staggered cycle). The “huge hollow” was $\sim 30\%$ metric artifact (the wall is clean, median dev 26°); CV is solved by K_V . The tube is *extrusion-driven* (pressure ~ 3): relaxation, Hertwig division, thickening, and emergent RD each remove or degrade it. The only genuinely low-stress route is *growth-driven* (monolayer, Okuda-quasi-static), but the first wiring *spiked* — likely because division still triggers on the *wedge* volume, not the monolayer’s $A \cdot h$, so activated cells *inflate* instead of *dividing*. Fixing the division trigger to the monolayer volume is the concrete next step toward a proliferation-built, low-stress tube.